The Political Economy of Public Debt and Electoral Polarization

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Abstract

This paper develops a politico-economic theory linking public debt and political polarization. We argue that fiscal constraints are politically asymmetric: debt disproportionately undermines left-wing electoral prospects by crowding out spending-intensive progressive agendas, inducing the right to strategically accumulate debt. Yet, strategic behavior is bounded: there is a party-specific debt threshold beyond which further debt accumulation is electorally unprofitable, with the right's threshold strictly lower than the left's. In equilibrium, a stochastic debt-electoral cycle emerges between these thresholds: left-wing governments accumulate debt, right-wing governments consolidate, and right-wing radicalism rises with indebtedness. The model offers a unified explanation for the U-shaped co-movement between public debt and right-wing radicalism, a pattern we document by constructing a new dataset of legislative elections for 31 OECD countries over the past century.

Keywords: Public debt, electoral cycles, political polarization
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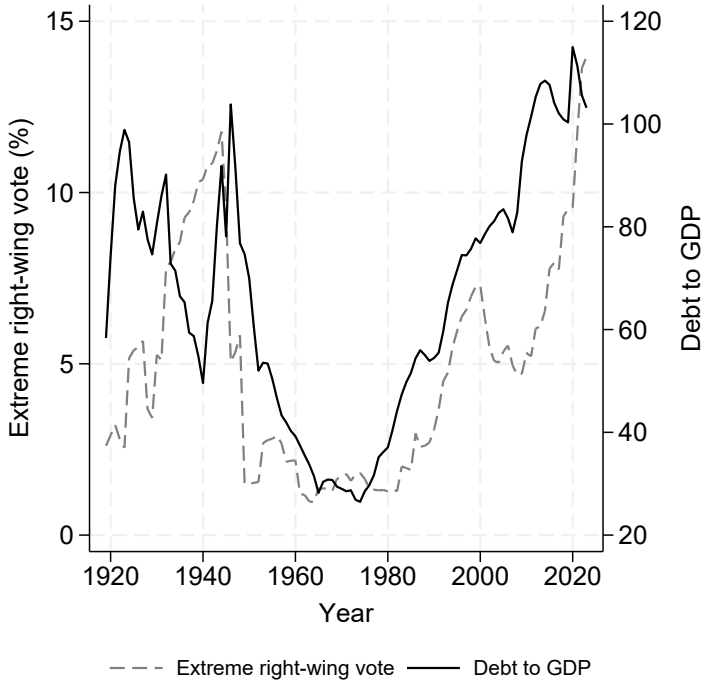
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1 Introduction

Over the past four decades, public debt has followed an upward trend in most advanced economies. However, this long-run trajectory is inconsistent with an efficient use of debt (Yared, 2019). The absence of a normative answer to this puzzle sparked the emergence of political theories of public debt, where debt overaccumulation arises from the present-bias and lack of commitment of policy-making. Among the amplifying forces responsible of debt overaccumulation, political polarization has gained particular relevance (Bouton et al., 2020; Funke et al., 2023). Yet, its origins and its interaction with fiscal and debt policies remain an open question. This paper closes this gap by developing a politico-economic theory connecting debt dynamics and political polarization, which rationalizes a century-long data pattern.

Figure 1: Extreme right vote share (%, left axis) and government debt-to-GDP (right axis), GDP-weighted means across 31 OECD countries, 1919–2023. Source: Authors' own elaboration.



To investigate the historical dynamics of public debt and political polarization, we begin by constructing a dataset on historical vote shares in legislative elections. Figure 1 documents a striking co-movement between public debt and extreme right-wing vote share (%) across 31 OECD countries over more than a century (GDP-weighted averages).[1] In the interwar period, public debt averaged 80–100 percent of GDP, while right-wing parties commanded 5–12 percent of total votes – culminating in the rise of fascist governments in Europe. The postwar decades brought a synchronized reversal: debt fell

[1]The sample includes: Australia, Austria, Belgium, Canada, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Japan, Latvia, Lithuania, Luxembourg, Netherlands, New Zealand, Norway, Poland, Portugal, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkey, and the United Kingdom. Our dataset substantially expands the country and time coverage of the data constructed by Funke et al. (2016). Appendix B provides details on data construction.

sharply to around 25 percent of GDP, and far right-wing vote shares contracted to historic lows near 1 percent through the 1970s. Since the early 1980s both series have risen in tandem. By the mid-2020s, average debt had returned to near its interwar peak, and far-right vote shares had surpassed those observed during Second World War.

The sustained rise in extreme right-wing parties' electoral support depicted in Figure 1 is consistent with recent anecdotal evidence. Examples include Trump in the United States (2016,2024), Orbán in Hungary (2010), Meloni in Italy (2022), Milei in Argentina (2023), Bolsonaro (2018) in Brazil, and more recently Kast in Chile (2025), all of whom campaigned on fiscal and debt restraint. Despite their electoral promises, public debt has continued to rise while the electoral support for these movements has only grown stronger. This raises the question, what explains the historical nexus between public debt and far-right polarization?

We address this question by developing an analytically tractable politico-economic model for the dynamic interplay between public debt and electoral uncertainty. The model consists of a small open economy with two-period lived overlapping generations. Citizens are born with heterogeneous labor productivity, they work and vote when young, and live off savings when old. In each period, two parties compete for office by announcing fiscal and debt plans to maximize their electoral probability. Public good provision is financed via an income tax and public debt. Parties differ only in their preference for the progressivity of public good provision—one party is left leaning and the other right-leaning. Crucially, the probability of winning depends on the fiscal and debt plans of both parties. Although parties maximize their current probability of winning, voters are forward-looking, so parties internalize the effects of debt issuance on their future electoral prospects.

Our model predicts that *fiscal constraints are politically asymmetric*: debt imposes a greater electoral cost on progressive platforms rather than conservative ones. The left's agenda—generous redistribution and progressive public good provision—is spending intensive and financed through debt issuance. However, inheriting large debts crowds out this agenda: servicing debt requires higher taxes and lower public spending, making left-wing promises undeliverable. In response, poorer voters do not shift right because they become more conservative, but because low-tax, low-spending agendas are simply more affordable. Debt is therefore a time bomb that gradually erodes the left's electoral support.

Although the right-wing narrative centers around tax cuts and fiscal restraint, issuing more debt serves a different strategic purpose: it constrains the fiscal space for left-wing agendas, reducing the probability that progressive ideas can ever be credibly delivered. This is not out of fiscal irresponsibility but about strategic fiscal manipulation. Moreover, when fiscal space is sufficiently tight, the right can have higher electoral gains by taking a more extreme position against public good provision.

Our theory predicts that strategic debt accumulation is bounded: there exists a party-specific debt threshold beyond which further debt accumulation is not electorally profitable. This debt threshold is lower for the right. Thus, as debt approaches to the right-wing's limit, endogenous political uncertainty gives rise to a stochastic debt-electoral cycle. When the left governs, it may raise debt above the right-wing's debt threshold. When the right returns to power, it decumulates debt, gradually restoring the fiscal space that strengthens the left's electoral prospects. The model thus predicts alternating periods of fiscal

consolidation and left-wing prevalence, followed by debt overaccumulation and the surge of right-wing radicalism. This mechanism offers a unified explanation for the U-shaped co-movement of public debt and right-wing radicalism observed over the last century.

This paper contributes to three strands of literature. The first strand develops politico-economic models where debt over-accumulation arises from policymakers' present bias and lack of commitment (Cukierman and Meltzer, 1989; Persson and Svensson, 1989; Alesina and Tabellini, 1990; Debortoli and Nunes, 2013). The literature has identified several forces that exacerbate debt accumulation, including an aging population (Arai et al., 2018; Song et al., 2012), voters' self-control problems (Bisin et al., 2015), declining reform efforts (Müller et al., 2019), increasing inequality (Arawatari and Ono, 2017), and rising political uncertainty (Battaglini and Coate, 2008; Yared, 2010; Bouton et al., 2020). The second strand documents how economic and financial shocks have favored the emergence of right-wing populism and political polarization (Doerr et al., 2022; Gyöngyösi and Verner, 2022; Ahlquist et al., 2020; Autor et al., 2020; Colantone and Stanig, 2019, 2018; Funke et al., 2016; Mian et al., 2014). The third strand empirically links political uncertainty, debt over-accumulation, and diminished economic activity (Reinhart and Rogoff, 2010; Reinhart et al., 2012; Baker et al., 2014, 2016).

This article adds to these three strands by developing a political theory where public debt has an endogenous electoral feedback. This mechanism gives rise to stochastic debt-electoral cycles that account for the historical co-movement of debt and right-wing radicalism, rationalizing the empirical link between fiscal distress and polarization. Overall, the model suggests that this link is not restricted to isolated episodes, but are instead cyclical in nature.

This paper is organized as follows. Section 2 presents the model. Section 3 describes the competitive equilibrium when policies are exogenous. Section 4 characterizes the political equilibrium. Section 5 studies the dynamics of public debt. Section 6 concludes.

2 The Model

Consider an overlapping generations model of a small open economy with two periods $t = 1,2$. The economy is populated at each period by two types of agents: citizens and politicians. Citizens consume two goods: a private good (c_t) and a domestic public good (g_t) provided by the government in charge. Politicians compete for office in each period by proposing fiscal and public debt schedules. Citizens vote for their most preferred candidate to maximize their discounted utilities.

2.1 Citizens

Citizens live for two periods: youth and old age. Each generation has a unit mass. When young, citizens first vote for their preferred politician.[2] Then, they work for a wage w_t, choose private consumption, and save to finance their consumption when old. Young agents are born with heterogeneous productivity $H \in [1, +\infty)$, distributed according to a Pareto distribution with shape parameter $\eta > 1$. The probability density function is $\gamma(H) = \frac{\eta}{H^{\eta+1}} \cdot \mathbb{1}_{H \geq 1}$, and the cumulative distribution function is $\Gamma(H) = (1 - H^{-\eta}) \cdot \mathbb{1}_{H \geq 1}$. Thus, the mean productivity is $\bar{H} = \frac{\eta}{\eta - 1}$.

In each period, both young and old agents receive public goods g_t. Public spending is financed by two sources: public debt issuance, b_{t+1}, and a proportional labor income tax, $\tau_t \in (0,1)$. The after-tax labor income of a young agent with productivity H is therefore $w_t H(1 - \tau_t)$.

2.2 Politicians

Two politicians $p \in \{D, R\}$—the Democrat (D) and the Republican (R)— compete for office . In each period t, they announce their proposed spending on public goods g_t, labor income tax τ_t, and public debt issuance b_{t+1}.

Politicians face uncertainty about their relative popularity. The Republican's relative popularity advantage is given by δ_t, where δ_t is uniformly distributed in $\left[-\frac{1}{2\psi}, \frac{1}{2\psi}\right]$, with $\psi > \frac{1}{2}$.[3] Politicians announce their platforms while uncertain about δ_t but knowing its probability distribution.

The Democrat and the Republican differ only in their preference over the progressivity (ρ) and intensity (θ) of public good provision. The Democrat has a stronger redistributive bias, so $\rho_D > \rho_R$ and $\theta_D > \theta_R$.

2.2.1 t=1

At $t = 1$, the inherited government debt is $b_1 > 0$. Politicians announce public good spending (g_1), tax rate (τ_1), and a future debt level (b_2) to maximize their probability of winning the election at $t = 1$. They commit to the announced policy if elected.

[2] To simplify the analysis, we study the properties of the model when only the young vote. The qualitative features of the model do not change when we account for the old.

[3] This general popularity measure augments the utility that any agent obtains when the Republican is elected. A similar approach is commonly used in probabilistic voting models (Lindbeck and Weibull, 1987; Persson and Tabellini, 2000).

2.2.2 t=2

At $t = 2$, politicians choose the tax rate (τ_2) and spending on their public good (g_2) to maximize their probability of winning at $t = 2$, while taking as given inherited debt, b_2. The government debt at the end of the second period is assumed to be zero ($b_3 = 0$).

2.2.3 Preferences

The expected discounted utility of a young citizen with productivity H, who consumes c_t during youth, c_{t+1} when old, given a public expenditure path (g_t, g_{t+1}) , is:

$$U_t^Y(H|p_t) = \mathbb{E}_t\left[\log(c_t) + \theta_t \log(g_t \cdot \phi(H|p_t)) + \beta\left(\log(c_{t+1}) + \theta_{t+1}\log\left(g_{t+1} \cdot \phi(H|p_{t+1})\right)\right)\right] \tag{2.1}$$

where $\phi(H|p_t)$ is the share of total public good spending allocated to a young individual with productivity H when a politician of type p_t is in power. Public good provision is progressive, i.e., $\phi'(H|p) < 0$. In particular, we assume that $\phi(H|p) = \frac{\bar{\phi}}{H^{\rho_p}}$, where $\bar{\phi}$ is such that $\int \phi(H)\gamma(H)\partial H = 1$ and $\rho_p > 0$ measures the progressivity of public good provision.[4]

The utility of an old agent who consumes c_t when government spending is g_t is:

$$U_t^O(H|p_t) = \mathbb{E}_t\left[\log(c_t) + \theta_t \log(g_t \cdot \phi(H|p_t))\right] \tag{2.2}$$

The expectation operators in equations (2.2) and (2.1) arise because there is uncertainty about who will be in power in the current and future period. As will be clear later, a distinguishing feature of the model is that electoral uncertainty is endogenous, as it depends on politicians' current proposals, which in turn impact future uncertainty.

3 Competitive Equilibrium

In this section, we study the competitive equilibrium under exogenous but uncertain fiscal and debt policies. In Section 3.1, we characterize citizens' optimal consumption-saving decisions. In Section 3.2, we determine electoral probabilities.

3.1 Consumption decisions

Citizens face a fixed gross return on savings $R = 1 + r$, a wage rate w_t, and a discount factor β. A young agent with productivity H chooses her consumption plans $(c_t(H), c_{t+1}(H))$ by solving:

$$\max_{c_t(H), c_{t+1}(H)} \left\{\mathbb{E}_t\left[\log(c_t(H)) + \beta\log(c_{t+1}(H))\right]\right\},$$
$$s.t.\;\; c_t(H) + \frac{c_{t+1}(H)}{R} = w_t(1-\tau_t)H,$$

[4] Specifically, $2 \cdot \int \phi(H|p)\gamma(H)\partial H = 1$ requires that $\bar{\phi} = \frac{\rho_p + \eta}{2\eta}$.

Since spending plans are chosen through voting, consumption plans are independent of (g_t, g_{t+1}). The first order condition of the problem is: $\mathbb{E}_t\left[\frac{1}{c_t(H)} - \beta R \frac{1}{c_{t+1}(H)}\right] = 0$. Thus, consumptions are given by:

$$c_t(H) = \frac{v_t H}{1+\beta}, \tag{3.1}$$

$$c_{t+1}(H) = \frac{\beta R v_t H}{1+\beta}, \tag{3.2}$$

where $v_t = w_t(1-\tau_t)$ is the after-tax lifetime income of a young agent with $H = 1$.

Omitting irrelevant constants, the expected discounted utility in period t of young and old citizens are:

$$U_t^Y(H) = \mathbb{E}_t\left[(1+\beta)\log(v_t \cdot H) + \theta_t \log(g_t \cdot \phi(H, p_t))\right] + \beta\,\mathbb{E}_t\left[\theta_{t+1}\log(g_{t+1}\cdot\phi(H,p_{t+1}))\right]\cdot \mathbb{1}_{t=1} + \mathbb{E}_t\left[\delta_t \cdot \mathbb{1}_{p_t=R}\right], \tag{3.3}$$

$$U_t^O(H) = \log(v_{t-1}\cdot H) + \mathbb{E}_t\left[\theta_t \log(g_t\cdot\phi(H,p_t))\right] + \mathbb{E}_t\left[\delta_t \cdot \mathbb{1}_{p_t=R}\right], \tag{3.4}$$

Since young agents do not survive to old age, the second expectation term in (3.3) disappears in period 2, thus $U_2^Y(H) = \mathbb{E}_1\left[(1+\beta)\log(v_2 H) + \theta_2 \log(g_2\cdot\phi(H,p_2)) + \tilde{\delta}_2 \cdot \mathbb{1}_{p_2=R}\right]$.

3.2 Politicians' electoral support

For any variable x, we define $\tilde{x} \equiv \frac{x}{1+\beta}$ throughout the paper to simplify notation. To characterize politicians' electoral support, it is useful to first identify the *swing voter*, defined as the least productive young agent who votes for the Republican. Formally, define the discounted lifetime utility of a young citizen born in t with productivity H under a politician of type p as follows:

$$\tilde{U}_t(H|p) = \log\left(v_t(p)H\right) + \tilde{\theta}_t \log\left(g_t(p)\cdot\bar{\phi}/H^{\rho_p}\right) + \beta\mathbb{E}_t\left[\tilde{\theta}_{t+1}\log\left(g_{t+1}(p,p_{t+1})\cdot\bar{\phi}/H^{\rho_{p_{t+1}}}\right)\right]\cdot\mathbb{1}_{t=1} + \tilde{\delta}_t\cdot\mathbb{1}_{p=R}, \tag{3.5}$$

where we have made explicit that the discounted lifetime utility $(v_t(p))$ and public good spending $(g_t(p))$ depend on the type of politician in power in t, $p_t = p$. Also, $g_{t+1}(p, p_{t+1})$ is the uncertain level of public spending, when a politician of type p_{t+1} is in power in the next period. Thus, future public spending depends on the history of governments in power (p_t, p_{t+1}). In Section 4, political turnover and the associated fiscal policy and debt accumulation are endogenized through elections.

A young citizen with productivity H votes for the Republican if:

$$\tilde{U}_t^Y\left(H|p_t = R\right) + \tilde{\delta}_t \geq \tilde{U}_t^Y\left(H|p_t = D\right) \tag{3.6}$$

The *swing voter*, $\hat{H}_t(\tilde{\delta}_t)$, is given by the productivity level such (3.6) holds with equality. Thus, the share of votes of the Democrat given δ_t is:

$$S_{D,t} = Pr\left[H \leq \hat{H}(\tilde{\delta}_t)\right] = \Gamma\left(\hat{H}(\tilde{\delta}_t)\right) = \max\left\{1 - \hat{H}(\tilde{\delta}_t)^{-\eta}, 0\right\} \tag{3.7}$$

Then, the probability that the Democrat wins the election is given by $\Pi_{D,t} = Pr\left[S_{D,t} > \frac{1}{2}\right]$. The following lemma characterizes the *swing voter* and the probability that the Democrat wins the election as a function of fiscal policies implemented by both politicians.

Lemma 1. *The swing voter is given by:*

$$\hat{H}_t(\tilde{\delta}_t) = \max\left\{\left(\frac{\bar{\phi}^{\tilde{\theta}_D - \tilde{\theta}_R}}{e^{\tilde{\delta}_t}} \cdot \frac{v_t(p_D)}{v_t(p_R)} \cdot \frac{g_t(p_D)^{\tilde{\theta}_D}}{g_t(p_R)^{\tilde{\theta}_R}} \cdot e^{\beta\mathbb{E}_t\left[\tilde{\theta}_{t+1}\log\left(\frac{g_{t+1}(p_D,p_{t+1})}{g_{t+1}(p_R,p_{t+1})}\right)\right]}\right)^{\frac{1}{\rho_D\tilde{\theta}_D - \rho_R\tilde{\theta}_R}}, 1\right\} \tag{3.8}$$

The probability that the Democrat wins the election is given by:

$$\begin{aligned}\Pi_{D,t} = \tilde{\Psi}\Bigg[&\frac{\rho_D\tilde{\theta}_D - \rho_R\tilde{\theta}_R}{\eta}\log\left(\frac{1}{2}\right) + (\tilde{\theta}_D - \tilde{\theta}_R)\log(\bar{\phi}) + \log\left(\frac{v_t(D)}{v_t(R)}\right)\\ &+ \log\left(\frac{g_t(D)^{\tilde{\theta}_D}}{g_t(R)^{\tilde{\theta}_R}}\right) + \beta\mathbb{E}_t\left[\tilde{\theta}_{t+1}\log\left(\frac{g_{t+1}(D,p_{t+1})}{g_{t+1}(R,p_{t+1})}\right)\right]\Bigg] + \frac{1}{2}\end{aligned} \tag{3.9}$$

The probability that the Republican wins the election is given by:

$$\begin{aligned}\Pi_{R,t} = \tilde{\Psi}\Bigg[&\frac{\rho_D\tilde{\theta}_D - \rho_R\tilde{\theta}_R}{\eta}\log(2) - (\tilde{\theta}_D - \tilde{\theta}_R)\log(\bar{\phi}) + \log\left(\frac{v_t(R)}{v_t(D)}\right)\\ &+ \log\left(\frac{g_t(R)^{\tilde{\theta}_R}}{g_t(D)^{\tilde{\theta}_D}}\right) + \beta\mathbb{E}_t\left[\tilde{\theta}_{t+1}\log\left(\frac{g_{t+1}(R,p_{t+1})}{g_{t+1}(D,p_{t+1})}\right)\right]\Bigg] + \frac{1}{2}\end{aligned} \tag{3.10}$$

Equations (3.9) and (3.10) show that the probability of winning the election is positively related to the politician's discounted utility and public good provision relative to those offered by her opponent. When choosing their political platform, politicians face a fundamental trade-off: increasing current public spending raises their probability of winning but generates a fiscal cost—either through higher current taxes or through higher debt—which translates into an electoral cost. In particular, increasing taxes reduces agents' lifetime utility, while issuing debt crowds out future public good provision via future tax increases.

The *swing voter* in equation (3.8) is constrained to $[1, +\infty)$, since the minimum productivity level is $H = 1$. When $\hat{H}_t(\tilde{\delta}_t) \in (1, +\infty)$, the *swing voter* is said to be "interior". Section 4.2.2 characterizes the conditions under which $\hat{H}_t(\tilde{\delta}_t)$ is interior.

4 Political Equilibrium

In this section, we study the political equilibrium under competitive elections between the Democrat and the Republican. In each period, a candidate of type p chooses fiscal and debt plans to maximize her electoral probability $\Pi_{p,t}$, subject to the government budget constraint:

$$\begin{aligned} &\max_{\tau_t, g_t, b_{t+1}} \Pi_{p,t} \\ &\quad s.t. \;\; g_t = b_{t+1} - Rb_t + \tau_t w_t \bar{H}, \end{aligned} \tag{4.1}$$

where b_t is the inherited debt from previous period. From equations (3.9) and (3.10), this problem is equivalent to solving:

$$\begin{aligned} &\max_{\tau_t(p), g_t(p), b_{t+1}(p)} \left\{ \log(v_t(p)) + \tilde{\theta}_p \log(g_t(p)) + \beta \mathbb{E}_t \left[\tilde{\theta}_{t+1} \log\left(g_{t+1}(p, p_{t+1}) \right) \right] \right\} \\ &\quad s.t. \;\; g_t(p) = b_{t+1}(p) - Rb_t + \tau_t(p) w_t \bar{H} \end{aligned} \tag{4.2}$$

In what follows, we characterize the solution to this problem recursively, starting from period 2. To simplify the analysis, we assume that the wage rate is fixed and given by $w > 0$. Importantly, when solving problem (4.2), politicians take future electoral probabilities as given. However, when characterizing the political equilibrium, we explicitly account for the endogenous effects of debt issuance on future electoral outcomes—requiring that politicians' beliefs about electoral probabilities are consistent with the equilibrium debt policy.

4.1 Politicians' fiscal and debt plans

In period 2, given an inherited debt b_2, a politician of type p_2 solves:[5]

$$\begin{aligned} &\max_{\tau_2(p), g_2(p)} \left\{ \log\left(v(p_2) \right) + \tilde{\theta}_p \log\left(g_2(p_2) \right) \right\} \\ &\quad s.t. \;\; g_2(p) = \tau_2(p) w \bar{H} - Rb_2, \end{aligned} \tag{4.3}$$

[5] The young agents in period 2 do not survive to old age, so the expectation term in equation (3.9) and (3.10) vanish.

In period 1, given an inherited debt b_1, a politician of type p_1 solves:[6]

$$\max_{\tau_1(p),g_1(p),b_2(p)} \left\{ \log(v(p_1)) + \tilde{\theta}_p \log(g(p)) + \beta \mathbb{E}_1 \left[\tilde{\theta}_2 \log(g(p,p_2)) \right] \right\}$$
$$s.t. \;\; g_1(p) = b_2(p) - Rb_1 + \tau_1(p) w\bar{H} \tag{4.4}$$

To simplify notation, we define the natural debt limit at period 2:[7]

$$\overline{b}_2 \equiv \frac{w\bar{H}}{R} \tag{4.5}$$

The following lemma characterizes the solutions to both problems:

Lemma 2. *Given an inherited public debt b_2, the tax rate ($\tau_2(p)$) and public good spending ($g_2(p)$) proposed by a type p politician at $t = 2$ are given by:*

$$\tau_2(p) = \frac{1}{1+\tilde{\theta}(p)} \left(\tilde{\theta}(p) + \frac{b_2}{\overline{b}_2} \right) \tag{4.6}$$
$$g_2(p) = \frac{R\tilde{\theta}(p)}{1+\tilde{\theta}(p)} \left(\overline{b}_2 - b_2 \right) \tag{4.7}$$

Given an inherited public debt b_1, public debt issuance ($b_2(p)$), the tax rate ($\tau_1(p)$), and public good spending ($g_1(p)$) proposed by a type p politician at $t = 1$ are given by:

$$b_2(p) = \frac{\overline{b}_2(1+\tilde{\theta}(p)) - \beta \mathbb{E}_1 \left[\tilde{\theta}_2 \right] \left(\overline{b}_2 - b_1 \right)}{1+\tilde{\theta}(p) + \beta \mathbb{E}_1 \left[\tilde{\theta}_2 \right]} \tag{4.8}$$
$$\tau_1(p) = 1 - \frac{1}{\beta \mathbb{E}_1 \left[\tilde{\theta}_2 \right]} \left(1 - \frac{b_2(p)}{\overline{b}_2} \right) \tag{4.9}$$
$$g_1(p) = \frac{\tilde{\theta}(p) \left(\overline{b}_2 - b_2(p) \right)}{\beta \mathbb{E}_1 \left[\tilde{\theta}_2 \right]} \tag{4.10}$$

Equation (4.8) captures a central tension of the model: (i) higher inherited debt b_1 constrains current fiscal space ($\downarrow (\overline{b}_2 - b_1)$), forcing the politician to issue more debt; and (ii) if the future government is expected to have a strong taste for public goods ($\uparrow \mathbb{E}_1 \left[\tilde{\theta}_2 \right]$), the politician issues less debt today. The second channel is in turn affected by debt issuance through its effect on electoral probabilities (3.9)–(3.10): higher debt issuance changes the probability of each type winning tomorrow, which in turn affects $\mathbb{E}_1 \left[\tilde{\theta}_2 \right]$. Equation (4.8) therefore constitutes a fixed point condition for debt issuance $b_2(p)$. In Section 5.1, we study in more detail how this channel operates for both parties.

[6] This is the general problem that politicians would solve in an infinite horizon model. Note the dependence on g_2. When the politician chooses b_2, she accounts for the future effect on expected public spending (g_2). This distinguishes the income tax τ_1 from debt b_2, which operate as different policy instruments. The tax rate today is chosen while trading-off the lifetime utility of the young against current public good provision. In contrast, the future debt is chosen while trading-off the future public spending decrease against current public spending increase. In sum, the income tax is used for intra-period redistribution, while debt is used for inter-period redistribution.

[7] The natural debt limit is obtained by replacing the maximum tax rate $\tau_2 = 1$ and $g_2 = 0$ in the government budget constraint in period 2.

4.2 Equilibrium electoral probabilities

In this section, we determine the equilibrium electoral probabilities at $t = 2$, which are required to characterize the dynamics of public debt in Section 5.3. To this end, in Section 4.2.1, we compute agents' equilibrium discounted utilities. In Section 4.2.2, we characterize the *swing voter*. Finally, in Section 4.2.3, we compute the equilibrium electoral probabilities at $t = 2$.

4.2.1 Discounted utility at $t = 2$

From equation (3.3), the normalized discounted utility of a young agent is given by:

$$\tilde{U}_2(H|p) = \log\left(v_2(p)H\right) + \tilde{\theta}_p \log\left(g_2(p)\frac{\bar{\phi}}{H^{\rho_p}}\right) + \mathbb{1}_{p=R}\tilde{\delta}_2$$

Use equations (4.6) and (4.7) and that $v_2(p) = (1-\tau_2(p))w$ to obtain:

$$\tilde{U}_2(H|p) = \log\left(w\frac{1}{\overline{b}_2(1+\tilde{\theta}_p)}\left[\overline{b}_2 - b_2\right]\right) + (1-\rho_p\tilde{\theta}_p)\log(H) + \tilde{\theta}_p \log\left(\frac{\tilde{\theta}_p}{1+\tilde{\theta}_p}R\left[\overline{b}_2 - b_2\right]\bar{\phi}\right) + \mathbb{1}_{p=R}\cdot\tilde{\delta}_2 \tag{4.11}$$

Note that,

$$\frac{\partial \tilde{U}_2(H|p)}{\partial H} = \frac{1-\rho_p\tilde{\theta}_p}{H} \tag{4.12}$$

Discounted utility $\tilde{U}_2(H|p)$ is strictly increasing in H if $\rho_p\tilde{\theta}_p < 1$, and is decreasing otherwise. Intuitively, if $\rho_p\tilde{\theta}_p < 1$, the politician has a low redistribute bias. Thus, she favors richer agents by charging lower taxes and spending less on the public good. The politicians' redistributive bias, $\rho_p\tilde{\theta}_p$, impacts the location of the *swing voter*, as shown in Lemma 3 below.

4.2.2 Swing voters at $t = 2$

Define $\omega(p) \equiv \log\left(w\frac{1}{\overline{b}_2(1+\tilde{\theta}_p)}\left[\overline{b}_2 - b_2\right]\right) + \tilde{\theta}_p\log\left(\frac{\tilde{\theta}_p}{1+\tilde{\theta}_p}R\left[\overline{b}_2 - b_2\right]\bar{\phi}\right)$, then equation (4.11) can be rewritten as:

$$\tilde{U}_2(H|p) = \omega(\tilde{\theta}_p) + (1-\rho_p\tilde{\theta}_p)\log(H) + \mathbb{1}_{p=R}\tilde{\delta}_2, \tag{4.13}$$

where $\omega(p)$ corresponds to the discounted utility provided by a politician of type p to the poorest agent in the economy (without accounting for general popularity). We have assumed that $\rho_D > \rho_R$ and $\theta_D > \theta_R$, which implies $1-\rho_R\tilde{\theta}_R > 1-\rho_D\tilde{\theta}_D$. Thus, when the politician's redistributive bias is sufficiently strong, $\rho_p\tilde{\theta}_p > 1$, the agent's discounted utility is strictly decreasing in H—richer agents receive lower utility. Conversely, if $\rho_p\tilde{\theta}_p < 1$, the politician has a regressive bias and poorer citizens receive lower utility. These properties have implications for the existence of an interior *swing voter*, as described in the following lemma:

Lemma 3. *Given a popularity level $\tilde{\delta}_2$, the swing voter, $\hat{H}_2(\tilde{\delta})$, is given by:*

1. *If $\omega(R) - \frac{1}{2\tilde{\Psi}} < \omega(D)$:*

$$\hat{H}_2(\tilde{\delta}) = \left(\alpha \cdot [\overline{b}_2 - b_2]^{\tilde{\theta}_D - \tilde{\theta}_R} \cdot e^{-\tilde{\delta}}\right)^{\frac{1}{\rho_D \tilde{\theta}_D - \rho_R \tilde{\theta}_R}}, \tag{4.14}$$

where $\alpha \equiv \frac{(1+\tilde{\theta}_R)^{1+\tilde{\theta}_R}}{(1+\tilde{\theta}_D)^{1+\tilde{\theta}_D}} \frac{\tilde{\theta}_D^{\tilde{\theta}_D}}{\tilde{\theta}_R^{\tilde{\theta}_R}} (R\bar{\phi})^{\tilde{\theta}_D - \tilde{\theta}_R}$.

2. *If $\omega(R) - \frac{1}{2\tilde{\Psi}} \geq \omega(D)$: $\hat{H}_2(\tilde{\delta}) = 1$ (Republican dominance).*

Lemma 3 shows that there is always a certain productivity level such that a young agent supports the Republican. In particular, when the Republican can guarantee a relatively high discounted utility for the poorest agent (case 1), she can win the election with certainty (*Republican dominance*). In contrast, a situation where the Democrat is elected with certainty is infeasible.

Consider case 2, where the *swing voter* is interior. Equation (4.14) shows that $\hat{H}_2(\tilde{\delta})$ is decreasing in inherited debt b_2. Intuitively, higher debt allows the Republican to attract more votes by promising lower taxes, at the expense of public spending. This result is not straightforward. From equation (4.13), higher debt b_2 shifts both $\tilde{U}_2(H|\tilde{\theta}_R)$ and $\tilde{U}_2(H|\tilde{\theta}_D)$ down. Thus, whether the intersection ($\hat{H}_2(\tilde{\delta})$) shifts right or left is in principle ambiguous. However, the Democrat's stronger bias on public good provision generates a larger downward shift in $\tilde{U}_2(H|\tilde{\theta}_D)$, pushing $\hat{H}$ downward, i.e., increasing the support for the Republican. Overall, the Republican's electoral advantage over the Democrat is greater when past governments have issued larger amounts of debt.

Lemma 3 leads to the following corollary:

Corollary 1. *The minimum debt level for a young agent with productivity H to support the Republican is given by:*

$$b_2(H,\delta) \equiv \overline{b}_2 - \left(\frac{1}{\alpha} H^{\rho_D \tilde{\theta}_D - \rho_R \tilde{\theta}_R} e^{\delta}\right)^{\frac{1}{\tilde{\theta}_D - \tilde{\theta}_R}} \tag{4.15}$$

Corollary 1 establishes that a young agent with productivity H votes for: (i) the Democrat if $b_2 \in [0, b_2(H,\delta))$, and (ii) the Republican if $b_2 \in [b_2(H,\delta), \overline{b}_2]$. The minimum debt level required to support the Republican ($b_2(H,\delta)$) is strictly decreasing in H. Intuitively, when the Democrat is elected, richer agents lose more when the inherited debt is larger, since a larger tax must be levied to finance the stronger redistribute bias of the Democrat.

4.2.3 Probability of winning the election at $t = 2$

Lemma 4. *Suppose that the swing voter is interior, $\omega(R) - \frac{1}{2\tilde{\Psi}} < \omega(D)$. The probability that the Democrat wins the election is strictly decreasing in inherited debt b_2:*

$$\frac{\partial \Pi_{D,2}}{\partial b_2} = -\frac{(\tilde{\theta}_D - \tilde{\theta}_R)}{\overline{b}_2 - b_2} < 0 \tag{4.16}$$

Lemma 4 establishes a key result of the paper: despite the Republican's distaste for debt, she enjoys a stronger electoral position if past governments have accumulated large amounts of debt. The marginal electoral benefit of inherited debt is higher when debt approaches its natural limit or when the Republican follows a more extreme political position relative to the Democrat (i.e., $\tilde{\theta}_D - \tilde{\theta}_R$ is large).

The expected value of public good provision intensity is given by $\mathbb{E}_1[\tilde{\theta}] = \tilde{\theta}_R + (\tilde{\theta}_D - \tilde{\theta}_R)\Pi_{D,2}$. Thus, from equation (4.16), we conclude that $\mathbb{E}_1[\tilde{\theta}]$ in strictly decreasing inherited debt b_2.

5 Public debt dynamics

In this section, we study the dynamics of public debt. We start by computing the effect of initial debt b_1 on future debt b_2. Then, we characterize the public debt dynamics arising from the stochastic political equilibrium.

5.1 Effect of current debt on debt issuance

Lemma 5. *The effect of current debt (b_1) on future debt (b_2) is given by:*

$$\frac{\partial b_2}{\partial b_1} = \frac{\beta R \mathbb{E}_1\left[\tilde{\theta}_2\right]\left(1+\tilde{\theta}_1+\beta\mathbb{E}_1\left[\tilde{\theta}_2\right]\right)}{\left(1+\tilde{\theta}_1+\beta\mathbb{E}_1\left[\tilde{\theta}_2\right]\right)^2 - \frac{\beta(\tilde{\theta}_D-\tilde{\theta}_R)^2(1+\tilde{\theta}_1)(2\bar{b}_2-b_1)}{\bar{b}_2-b_2}} > 0 \tag{5.1}$$

Equation (5.1) captures the key role of political uncertainty on debt accumulation. To interpret this result, consider the first order condition of b_2 from problem (4.4):

$$\underbrace{\frac{\tilde{\theta}(p)}{g_1(p)}}_{\text{Marginal benefit of } g_1} = \underbrace{\beta R \mathbb{E}_1\left[\frac{\tilde{\theta}_2}{g_2}\frac{\tilde{\theta}_2}{1+\tilde{\theta}_2}\right]}_{\text{Future marginal cost on } g_2} \tag{5.2}$$

Condition (5.2) states that a period 1-politician of type p chooses debt issuance b_2 by equating the marginal benefit of current public good spending to the expected marginal cost of b_2 on future public spending. This is the standard trade-off present in positive theories of public debt (e.g, see Arawatari and Ono, 2017; Song et al., 2012). The new additional force affecting debt accumulation in our model is captured by the second term in the denominator of (5.1), which arises from the effect of b_2 on electoral probabilities.

5.1.1 The baseline case without endogenous political uncertainty

To interpret (5.1), first consider the case in which politicians do not internalize the effect of debt on political uncertainty, as in the baseline models in the literature. Thus, the equation (5.1) becomes:

$$\frac{\partial b_2}{\partial b_1} = \frac{\beta R \mathbb{E}_1\left[\tilde{\theta}_2\right]}{1+\tilde{\theta}_1+\beta\mathbb{E}_1\left[\tilde{\theta}_2\right]} > 0 \tag{5.3}$$

The intuition is as follows. Higher inherited debt b_1 tightens the government budget constraint, reducing current public good provision g_1 and thus increasing its marginal benefit (*direct budgetary effect*). To satisfy (5.2), the politician issues more debt, which increases g_1 while reducing expected future public good provision g_2, thereby effectively restoring (5.2).

Equation (5.1) reveals another important property regarding debt accumulation: higher discounted expected future value of public good provision, $\beta R\mathbb{E}_1[\tilde{\theta}_2]$, increases the rate of debt accumulation (*Intertemporal political trade-off*). Moreover, this debt increase is larger for politicians with a significantly lower current taste for public good provision, $\tilde{\theta}_1$. Formally,

$$\frac{\partial^3 b_2}{\partial b_1 \partial \mathbb{E}_1[\tilde{\theta}_2] \partial \tilde{\theta}_1} = -\frac{\beta R\left(1+\tilde{\theta}_1-\beta R\mathbb{E}_1[\tilde{\theta}_2]\right)}{\left(1+\tilde{\theta}_1+\beta R\mathbb{E}_1\left[\tilde{\theta}_2\right]\right)^3} \tag{5.4}$$

Suppose that the current party in power is Republican. Expression (5.4) is positive when the Republican distaste for public good provision is strong, i.e., when $\tilde{\theta}_1 = \tilde{\theta}_R$ is low. Intuitively, higher $\mathbb{E}_1[\tilde{\theta}_2]$ means that the probability that the Democrat wins in the future ($\Pi_{D,2}$) is larger. Hence, the Republican issues more debt to discipline the future Democrat's public spending agenda. This is the classic strategic motive in positive theories of public debt (Persson and Svensson, 1989; Alesina and Tabellini, 1990). However, there is a key difference of our model compared to the literature: the strategic incentives through which politicians over-accumulative debt are partisan-specific.

When the Democrat is currently in power and has a strong redistributive bias, i.e., $\tilde{\theta}_1 = \tilde{\theta}_D$ is high, expression (5.4) becomes negative. Thus, the debt increase is lower relative to the Republican. Intuitively, since a high $\mathbb{E}_1[\tilde{\theta}_2]$ implies that the Democrat is likely to remain in power in the future, she does not accumulate too much debt to avoid inheriting excessive debt that would constraint her future agenda.

Why does the Democrat still issue debt when she is likely to remain in power in the future? A higher future expected value of public good provision ($\mathbb{E}_1[\tilde{\theta}_2]$) increases future public good spending (g_2), which decreases the marginal cost of debt in the first order condition (5.2). In response, the politician in power issues more debt to increase g_1, thereby reducing its current marginal benefit and restoring optimality.

Overall, the same increase in $\mathbb{E}_1[\tilde{\theta}_2]$, driven by a higher future electoral probability of the Democrat ($\Pi_{D,2}$), generates opposite strategic incentives across politicians. The Republican issues more debt to constrain the future Democrat's agenda, while the Democrat moderates debt accumulation to preserve fiscal space for her own future agenda. The disagreement in debt accumulation between both parties is larger the greater the differences in their public spending preferences, $\tilde{\theta}_D - \tilde{\theta}_R$.

5.1.2 The role of endogenous political uncertainty

To understand the role of endogenous political uncertainty on debt accumulation, rewrite expression (5.1) as follows:

$$\frac{\partial b_2}{\partial b_1} = \frac{\overbrace{\beta R \mathbb{E}_1\left[\tilde{\theta}_2\right]}^{\text{Direct budgetary effect}}}{\underbrace{\left(1+\tilde{\theta}_1+\beta\mathbb{E}_1\left[\tilde{\theta}_2\right]\right)}_{\text{Intertemporal political trade-off}} + \underbrace{\frac{\boldsymbol{\beta}\frac{\partial \mathbb{E}_1[\tilde{\boldsymbol{\theta}}_2]}{\partial \boldsymbol{b}_2}(\mathbf{1}+\tilde{\boldsymbol{\theta}}_1)(2\overline{\boldsymbol{b}}_2-\boldsymbol{b}_1)}{\left(\mathbf{1}+\tilde{\boldsymbol{\theta}}_1+\boldsymbol{\beta}\mathbb{E}_1\left[\tilde{\boldsymbol{\theta}}_2\right]\right)}}_{\text{Electoral feedback effect}}}, \tag{5.5}$$

where the bold symbol corresponds to the second negative term in equation (5.1), which captures the general equilibrium effect of debt issuance on future electoral probabilities (*electoral feedback effect*).

From Lemma 4, higher debt issuance ($\uparrow b_2$) decreases the probability that the Democrat wins the future election. Thus, the expected future redistribute bias $\mathbb{E}_1[\tilde{\theta}_2]$ decreases. As a result, the future marginal cost of debt issuance declines, motivating politicians to increase debt issuance to finance more current public spending. This debt increase in turn amplifies the electoral feedback effect inducing more debt accumulation.

Overall, when political uncertainty responds to debt issuance, public debt increases at a faster rate than under the baseline case, without an electoral feedback. The higher is polarization ($\tilde{\theta}_D - \tilde{\theta}_R$) or the stronger the redistribute bias of the current administration ($\tilde{\theta}_1$), the larger is the electoral feedback effect on debt accumulation.

5.2 Infinite periods model

In order to study the joint dynamics of public debt and political uncertainty, we consider the baseline model with infinite periods, $t = \{1,2,...\}$. In each period, both politicians compete for office and chose their fiscal plans to solve:

$$\begin{aligned} \max_{\tau_t(p),g_t(p),b_{t+1}(p)} \Pi_{p,t} \Leftrightarrow \max_{\tau_t(p),g_t(p),b_{t+1}(p)} &\left\{\log(v_t(p)) + \tilde{\theta}_p \log(g_t(p)) + \beta\mathbb{E}_t\left[\tilde{\theta}_{t+1}\log\left(g_{t+1}(p,p_{t+1})\right)\right]\right\} \\ s.t \quad & g_t(p) = b_{t+1}(p) - Rb_t + \tau_t(p)\, w_t \bar{H} \end{aligned} \tag{5.6}$$

As in Song et al. (2012), we restrict our analysis to Markov-perfect equilibria (MPE). Thus,the main payoff-relevant state variables are current debt b_t and the identity of the party who proposes a fiscal plan p_t.

Definition 1. *An MPE is a triple* (B,G,T)*, where* $B:[0,\overline{b}]\times\{D,R\}\to[0,\overline{b}]$ *is a debt rule,* $b_{t+1}=B(b_t,w_t,p_t)$*,* $G:[0,\overline{b}]\times\{D,R\}\to[0,\overline{g}]$ *is a public spending rule,* $g_t=G(b_t,w_t,p_t)$*, and* $T:[0,\overline{b}]\times\{D,R\}\to[0,1]$ *is a tax rule such that:*

1. $B(b_t,p_t),G(b_t,p_t),T(b_t,p_t)$ *solve problem* (5.6).

2. *The stochastic transitions from p_t to $p_{t+1} = D$ and $p_{t+1} = R$ are determined by equations* (3.9) *and* (3.10)*, respectively, where the transition probabilities are consistent with the equilibrium policy rules* (B, G, T).

In Definition 1 we have denoted the natural debt and spending limits by: $\overline{b} = \frac{w_t \bar{H}}{R-1}$ and $\bar{g} = \frac{Rw\bar{H}}{R-1}$, respectively. Unlike the standard MPE concepts in the political economy models of public debt (Battaglini and Coate, 2008; Song et al., 2012; Arawatari and Ono, 2017), our model requires the endogenous transition probabilities to be consistent with the equilibrium policy rules (B, G, T). The policy functions of debt, public spending, and taxes inherit an analogous functional form to those of Lemma 2 in the two-periods model. In what follows, we denote the policy function of debt, taxes and spending of type-p party at t by $b_{t+1}(b_t|p)$, $\tau(b_t|p_t)$, and $g(b_t|p)$, respectively.

Lemma 6. *In a stationary Markov-perfect equilibrium, the policy functions of the type-p party are*

$$b_{t+1}(b_t \mid p) = \bar{b} - \frac{R\beta \mathbb{E}_t[\tilde{\theta}_{t+1}]\left(\bar{b} - b_t\right)}{1 + \tilde{\theta}_p + \beta \mathbb{E}_t[\tilde{\theta}_{t+1}]}, \tag{5.7}$$

$$\tau(b_t \mid p) = 1 - \frac{1}{(R-1)\beta \mathbb{E}_t[\tilde{\theta}_{t+1}]}\left(1 - \frac{b_{t+1}(b_t \mid p)}{\bar{b}}\right), \tag{5.8}$$

$$g(b_t \mid p) = \frac{R\tilde{\theta}_p\left(\bar{b} - b_t\right)}{1 + \tilde{\theta}_p + \beta \mathbb{E}_t[\tilde{\theta}_{t+1}]}. \tag{5.9}$$

5.3 Stochastic equilibrium dynamics: Public debt and electoral cycles

The infinite period version of the model gives rise to the following stochastic political equilibrium:

Proposition 1.

1. *There exists a unique party-specific stationary public debt level, $b_{t+1}(b_p^*|p) = b_p^*$, with $b_D^* > b_R^*$.*

2. *The transition dynamics of public debt and electoral uncertainty are given by:*

   (a) *If $b_t < b_R^*$, then b_{t+1} follows a stochastic increasing path, where $b_{t+1} = b_{t+1}(b_t|D)$ with probability $\Pi_{D,t}$ and $b_{t+1} = b_{t+1}(b_t|R)$ with probability $\Pi_{R,t}$. Moreover, $\Pi_{R,t}$ increases over time as debt approaches to b_R^*.*

   (b) *If $b_t \in [b_R^*, b_D^*]$, public debt attains a stochastic-cycling equilibrium, where b_t stays in $[b_R^*, b_D^*]$ forever. Debt increases to $b_{t+1} = b_{t+1}(b_t|D)$ with probability $\Pi_{D,t}$, while it decreases to $b_{t+1} = b_{t+1}(b_t|R)$ with probability $\Pi_{R,t}$. Moreover, $\Pi_{R,t}$ increases when debt approaches to b_D^*, while $\Pi_{R,t}$ decreases when debt approaches to b_R^*.*

   (c) *If $b_t > b_D^*$, then b_{t+1} follows a stochastic decreasing path, where $b_{t+1} = b_{t+1}(b_t|D)$ with probability $\Pi_{D,t}$ and $b_{t+1} = b_{t+1}(b_t|R)$ with probability $\Pi_{R,t}$. Moreover, $\Pi_{R,t}$ decreases over time as debt approaches to b_R^*.*

Proposition 1 shows that the stochastic behavior of public debt and elections is characterized across three debt ranges: (i) $b_t < b_R^*$, (ii), $b_t \in [b_R^*, b_D^*]$, and (iii) $b_t > b_D^*$, where b_p^* corresponds to the stationary debt limit of party p. Figure 2 illustrates the proposition.

First, if $b_t < b_R^*$, both parties accumulate debt over time. As debt approaches b_R^*, the probability that the Republican is elected increases, since debt constrains the fiscal space for the progressive Democrat's agenda.

Second, if $b_t \in [b_R^*, b_D^*]$, then debt enters in stochastic cycles. Whenever the Republican is elected, she reduces debt, as the current level exceeds the threshold at which further accumulation is electorally profitable. Paradoxically, when the Republican decreases debt, she ultimately raises the probability that the Democrat is elected by giving fiscal space for left-wing progressive agendas. Whenever the Democrat is elected, she accumulates debt to finance her spending-intensive agenda. However, as the fiscal space narrows, the probability that the Republican wins the election increases, reversing the upward trend of debt. Moreover, in periods where the Democrat accumulates debt, the Republican can raise her electoral gain by taking a more extreme position against public good provision, i.e., by lowering $\tilde{\theta}_R$.

Third, if $b_t > b_D^*$, then both parties are in the range where accumulating additional debt is not electorally profitable. Thus, debt decumulates until it falls into the stochastic cycling equilibrium.

Overall, the model predicts that eventually public debt converges to a stochastic-cycling equilibrium. Economies alternate between periods of fiscal consolidation and left-wing prevalence, followed by debt overaccumulation and the surge of right-wing radicalism. This mechanism offers a unified rationale for the U-shaped pattern of public debt and right-wing radicalism presented in Figure 1. Moreover, the model predicts that countries will eventually transition to consolidation phase of the cycle. Thus, public debt will decumulate, leading to the prevalence of left-wing agendas and the fall of right-wing radicalism.

Figure 3 presents a simulated paths for public debt (solid line) and Republican party's electoral probability (dashed line), each one corresponding to the average of 1,000 simulations over ten periods. As implied by item b) of Proposition 1, the stochastic political equilibrium generates a co-movement between debt and the right-wing electoral support. This pattern is consistent with the data shown in Figure 1.

Figure 2: Stochastic public debt equilibrium

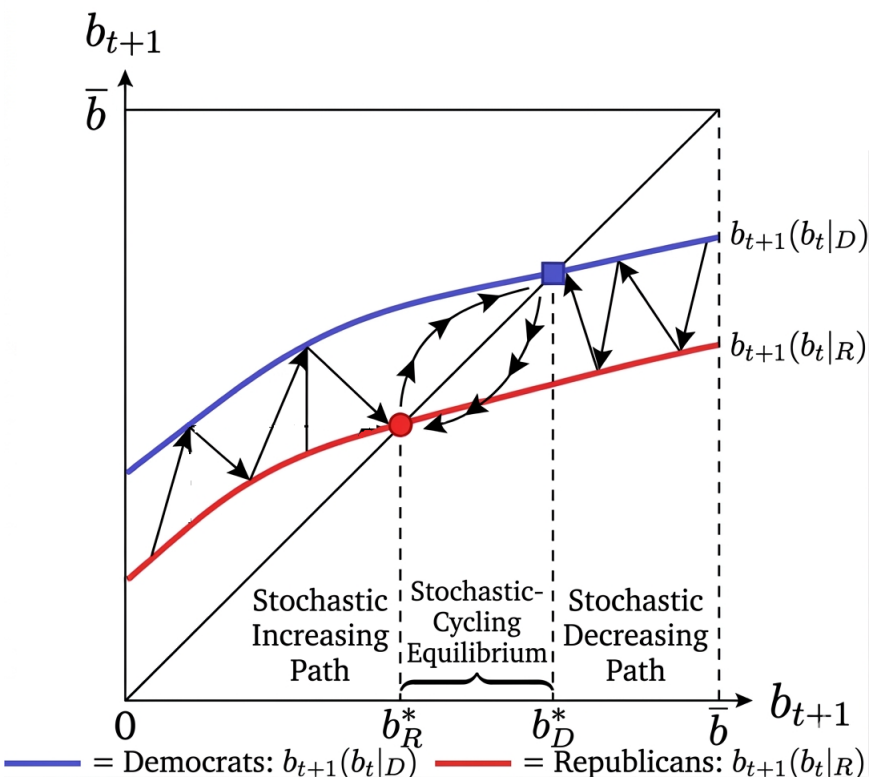
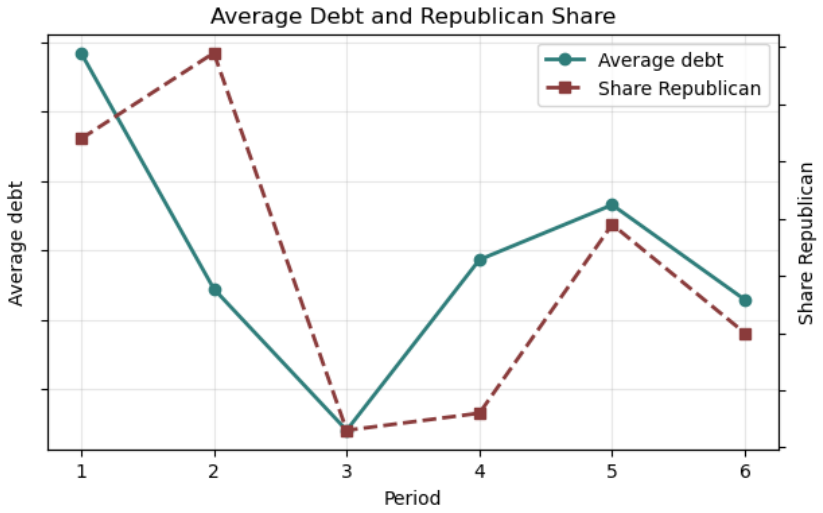


Figure 3: Simulated public debt and electoral cycles



6 Conclusions

We begin by constructing historical data on legislative vote shares for 31 OECD countries over more than a century (1919-2023). The data reveals a striking U-shaped co-movement between public debt and extreme right-wing vote share. This paper develops a tractable politico-economic theory that rationalizes the observed joint behavior of public debt and far-right political polarization.

The central theoretical argument is that *fiscal constraints are politically asymmetric*: debt imposes a greater electoral cost on progressive platforms rather than conservative ones, inducing the right to strategically accumulate debt to hurt the left's electoral prospects. As debt accumulates, the right can have higher electoral gains through a more extreme agenda against public good provision. However, strategic behavior is bounded by a party-specific debt threshold beyond which debt accumulation is electorally unprofitable, with the right's threshold strictly lower than the left's. As a result, a stochastic debt-electoral cycle emerges between these two thresholds: when in power, the left accumulates debt, the right consolidates, and right-wing radicalism rises with debt. This mechanism rationalizes the U-shaped co-movement between debt and right-wing radicalism observed in the last century.

This paper opens the door for several future research avenues. First, given the evidence on the economic cost of debt and political polarization, understanding the mechanisms through which these measures reinforce each other is key to design institutions that can contain them. In particular, given the mixed record of fiscal rules in controlling debt accumulation, an important open question concerns the design of political systems that can more effectively prevent debt-polarization spirals.

Second, our theory lays the foundation for further research on the endogenous feedback between policies, inequality, and economic outcomes. In particular, redistributive policies may break or reinforce the debt-polarization cycle identified in this paper through the changes they induce on wealth and income inequality.

Third, incorporating into our framework the interaction between an independent central bank and a politically polarized fiscal authority could shed light on the fundamental forces driving fiscal and monetary dominance across countries.

Finally, extending the framework to open economies can introduce additional amplification mechanisms: rising far-right support may increase sovereign risk premia, further constraining fiscal space and fueling voter resentment. This extension seems particularly promising given the sovereign debt crisis experiences in Greece, Italy, and Spain, where IMF adjustment programs and the European Union Stability Growth Pact interacted forcefully with domestic political dynamics.

A Proofs

A.1 Proof of Lemma 1

Lemma 1. *The swing voter is given by:*

$$\hat{H}_t(\tilde{\delta}_t) = \left(\frac{\bar{\phi}^{\tilde{\theta}_D - \tilde{\theta}_R}}{e^{\tilde{\delta}_t}} \cdot \frac{v_t(p_D)}{v_t(p_R)} \cdot \frac{g_t(p_D)^{\tilde{\theta}_D}}{g_t(p_R)^{\tilde{\theta}_R}} \cdot e^{\beta \mathbb{E}_t \left[\tilde{\theta}_{t+1} \log\left(\frac{g_{t+1}(p_D, p_{t+1})}{g_{t+1}(p_R, p_{t+1})} \right) \right]} \right)^{\frac{1}{\rho_D \tilde{\theta}_D - \rho_R \tilde{\theta}_R}} \tag{A.1}$$

The probability that the Democrat wins the election is given by:

$$\begin{aligned} \Pi_{D,t} = \tilde{\Psi} \Bigg[& \frac{\rho_D \tilde{\theta}_D - \rho_R \tilde{\theta}_R}{\eta} \log\left(\frac{1}{2}\right) + (\tilde{\theta}_D - \tilde{\theta}_R) \log(\bar{\phi}) + \log\left(\frac{v_t(D)}{v_t(R)} \right) \\ & + \log\left(\frac{g_t(D)^{\tilde{\theta}_D}}{g_t(R)^{\tilde{\theta}_R}} \right) + \beta \mathbb{E}_t \left[\tilde{\theta}_{t+1} \log\left(\frac{g_{t+1}(D, p_{t+1})}{g_{t+1}(R, p_{t+1})} \right) \right] \Bigg] + \frac{1}{2} \end{aligned} \tag{A.2}$$

The probability that the Republican wins the election is given by:

$$\begin{aligned} \Pi_{R,t} = \tilde{\Psi} \Bigg[& \frac{\rho_D \tilde{\theta}_D - \rho_R \tilde{\theta}_R}{\eta} \log(2) - (\tilde{\theta}_D - \tilde{\theta}_R) \log(\bar{\phi}) + \log\left(\frac{v_t(R)}{v_t(D)} \right) \\ & + \log\left(\frac{g_t(R)^{\tilde{\theta}_R}}{g_t(D)^{\tilde{\theta}_D}} \right) + \beta \mathbb{E}_t \left[\tilde{\theta}_{t+1} \log\left(\frac{g_{t+1}(R, p_{t+1})}{g_{t+1}(D, p_{t+1})} \right) \right] \Bigg] + \frac{1}{2} \end{aligned} \tag{A.3}$$

Proof. The *swing voter*, $\hat{H}_t(\tilde{\delta}_t)$, is such that (3.6) hold with equality. Thus,

$$\begin{aligned} \tilde{U}^Y \left(\hat{H}_t | p_t = p_D \right) &= \tilde{U}^Y \left(\hat{H}_t | p_t = p_R \right) + \tilde{\delta}_t, \\ &\Leftrightarrow \log(v_t(p_D) \hat{H}_t) + \tilde{\theta}_D \log(g_t(p_D) \bar{\phi} / \hat{H}_t^{\rho_D}) + \beta \mathbb{E}_t \left[\tilde{\theta}_{t+1} \log(g_{t+1}(p_D, p_{t+1})) \right] \\ &\quad = \log(v_t(p_R) \hat{H}_t) + \tilde{\theta}_R \log(g_t(p_R) \bar{\phi} / \hat{H}_t^{\rho_R}) + \beta \mathbb{E}_t \left[\tilde{\theta}_{t+1} \log(g_{t+1}(p_R, p_{t+1})) \right] + \tilde{\delta}_t, \\ &\Leftrightarrow \hat{H}_t = \left(\frac{\bar{\phi}^{\tilde{\theta}_D - \tilde{\theta}_R}}{e^{\tilde{\delta}}} \cdot \frac{v_t(p_D)}{v_t(p_R)} \cdot \frac{g_t(p_D)^{\tilde{\theta}_D}}{g_t(p_R)^{\tilde{\theta}_R}} \cdot e^{\beta \mathbb{E}_t \left[\tilde{\theta}_{t+1} \log\left(\frac{g_{t+1}(p_D, p_{t+1})}{g_{t+1}(p_R, p_{t+1})} \right) \right]} \right)^{\frac{1}{\rho_D \tilde{\theta}_D - \rho_R \tilde{\theta}_R}}, \end{aligned} \tag{A.4}$$

which corresponds to expression (A.1). The probability that the Democrat wins the election is given by:

$$
\begin{aligned}
\Pi_{D,t} &= Pr\left[S_{D,t}\left(\tilde{\delta}_t\right) \geq \frac{1}{2}\right] \\
&= Pr\left[\left(\hat{H}_t(\tilde{\delta}_t)\right)^{-\eta} \leq \frac{1}{2}\right] = Pr\left[\frac{e^{\tilde{\delta}_t}}{\bar{\phi}^{\tilde{\theta}_D-\tilde{\theta}_R}} \cdot \frac{v(p_R)}{v(p_D)} \cdot \frac{g(p_R)^{\tilde{\theta}_R}}{g(p_D)^{\tilde{\theta}_D}} \cdot e^{-\beta\mathbb{E}_t\left[\tilde{\theta}_{t+1}\log\left(\frac{g_{t+1}(p_D,p_{t+1})}{g_{t+1}(p_R,p_{t+1})}\right)\right]} \leq \left(\frac{1}{2}\right)^{\frac{\rho_D\tilde{\theta}_D-\rho_R\tilde{\theta}_R}{\eta}}\right] \\
&= Pr\left[e^{\tilde{\delta}_t} \leq \left(\frac{1}{2}\right)^{\frac{\rho_D\tilde{\theta}_D-\rho_R\tilde{\theta}_R}{\eta}} \bar{\phi}^{\tilde{\theta}_D-\tilde{\theta}_R} \cdot \frac{v(p_D)}{v(p_R)} \cdot \frac{g(p_D)^{\tilde{\theta}_D}}{g(p_R)^{\tilde{\theta}_R}} \cdot e^{\beta\mathbb{E}_t\left[\tilde{\theta}_{t+1}\log\left(\frac{g_{t+1}(p_D,p_{t+1})}{g_{t+1}(p_R,p_{t+1})}\right)\right]}\right] \\
&= Pr\Big[\tilde{\delta}_t \leq \frac{\rho_D\tilde{\theta}_D-\rho_R\tilde{\theta}_R}{\eta}\log\left(\frac{1}{2}\right) + (\tilde{\theta}_D-\tilde{\theta}_R)\log(\bar{\phi}) + [\log(v(p_D)) - \log(v(p_R))] \\
&\quad + \tilde{\theta}_D\log(g(p_D)) - \tilde{\theta}_R\log(g(p_R)) + \beta\mathbb{E}_t\left[\tilde{\theta}_{t+1}\log\left(\frac{g_{t+1}(p_D,p_{t+1})}{g_{t+1}(p_R,p_{t+1})}\right)\right]\Big] \\
&= \tilde{\Psi}\left[\frac{\rho_D\tilde{\theta}_D-\rho_R\tilde{\theta}_R}{\eta}\log\left(\frac{1}{2}\right) + (\tilde{\theta}_D-\tilde{\theta}_R)\log(\bar{\phi}) + [\log(v(p_D)) - \log(v(p_R))]\right. \\
&\quad \left. + \tilde{\theta}_D\log(g(p_D)) - \tilde{\theta}_R\log(g(p_R)) + \beta\mathbb{E}_t\left[\tilde{\theta}_{t+1}\log\left(\frac{g_{t+1}(p_D,p_{t+1})}{g_{t+1}(p_R,p_{t+1})}\right)\right]\right] + \frac{1}{2},
\end{aligned}
$$

which corresponds to expression (A.2). The probability that the Republican wins the election is given by $\Pi_{R,t} = 1 - \Pi_{D,t}$, which leads to expression (A.3). □

A.2 Proof of Lemma 2

Lemma 2. *Given an inherited public debt b_2, the tax rate ($\tau_2(p)$) and public good spending ($g_2(p)$) proposed by a type p politician at $t = 2$ are given by:*

$$\tau_2(p) = \frac{1}{1+\tilde{\theta}(p)}\left(\tilde{\theta}(p) + \frac{b_2}{\overline{b}_2}\right) \tag{A.5}$$

$$g_2(p) = \frac{R\tilde{\theta}(p)}{1+\tilde{\theta}(p)}\left(\overline{b}_2 - b_2\right) \tag{A.6}$$

Given an inherited public debt b_1, public debt issuance ($b_2(p)$), the tax rate ($\tau_1(p)$), and public good spending ($g_1(p)$) proposed by a type p politician at $t = 1$ are given by:

$$b_2(p) = \frac{\overline{b}_2(1+\tilde{\theta}(p)) - \beta\mathbb{E}_1\left[\tilde{\theta}_2\right]\left(\overline{b}_2 - b_1\right)}{1+\tilde{\theta}(p)+\beta\mathbb{E}_1\left[\tilde{\theta}_2\right]} \tag{A.7}$$

$$\tau_1(p) = 1 - \frac{1}{\beta\mathbb{E}_1\left[\tilde{\theta}_2\right]}\left(1 - \frac{b_2(p)}{\overline{b}_2}\right) \tag{A.8}$$

$$g_1(p) = \frac{\tilde{\theta}(p)\left(\overline{b}_2 - b_2(p)\right)}{\beta\mathbb{E}_1\left[\tilde{\theta}_2\right]} \tag{A.9}$$

Proof. To simplify notation we omit the dependence of all variables on the politician in power, p. The

first order condition of problem (4.3) gives:

$$\frac{1}{v_2}\frac{\partial v_2}{\partial \tau_2}+\frac{\tilde{\theta}_2}{g_2}\frac{\partial g_2}{\partial \tau_2}=0.$$

Using that $\frac{\partial v_2}{\partial \tau_2}=-w$ and that $\frac{\partial g_2}{\partial \tau_2}=w\bar{H}$, we obtain that:

$$g_2=\tilde{\theta}_2(1-\tau_2)w\bar{H} \tag{A.10}$$

Replacing in the budget constraint of the government gives the tax rate:

$$\begin{aligned}\tilde{\theta}_2 w\bar{H}(1-\tau_2)&=\tau_2 w\bar{H}-Rb_2\\ \Rightarrow \tau_2&=\frac{\tilde{\theta}_2}{1+\tilde{\theta}_2}+\frac{Rb_2}{(1+\tilde{\theta}_2)w\bar{H}}\end{aligned} \tag{A.11}$$

Replacing (A.11) into (A.10) gives the expression for g_2:

$$g_2=\frac{\tilde{\theta}_2}{1+\tilde{\theta}_2}\left(w\bar{H}-Rb_2\right) \tag{A.12}$$

Expressing equations (A.11) and (A.12) in terms of the natural debt limit leads to equations (A.5) and (A.6). The FOC of τ_1 is given by:

$$\begin{aligned}\frac{1}{v}\frac{\partial v}{\partial \tau_1}+\frac{\tilde{\theta}_1}{g_1}\frac{\partial g_1}{\partial \tau_1}=0 \Leftrightarrow \frac{1}{v}&=\frac{\tilde{\theta}_1}{g_1}\bar{H}\\ \Rightarrow g_1&=\tilde{\theta}_1(1-\tau_1)w\bar{H}\end{aligned} \tag{A.13}$$

The FOC of b_2 is given by:

$$\begin{aligned}\frac{\tilde{\theta}_1}{g_1}\frac{\partial g_1}{\partial b_2}+\beta\mathbb{E}_1\left[\frac{\tilde{\theta}_2}{g_2}\frac{\partial g_2}{\partial b_2}\right]&=0\\ \Leftrightarrow \frac{\tilde{\theta}_1}{g_1}&=\beta R\mathbb{E}_1\left[\frac{\tilde{\theta}_2}{g_2}\frac{\tilde{\theta}_2}{1+\tilde{\theta}_2}\right]\\ \Rightarrow g_1&=\frac{1}{\beta R\mathbb{E}_1\left[\frac{\tilde{\theta}_2}{\tilde{\theta}_1 g_2}\frac{\tilde{\theta}_2}{1+\tilde{\theta}_2}\right]}\end{aligned} \tag{A.14}$$

Replacing (A.12) into (A.14) gives:

$$g_1=\frac{\tilde{\theta}_1}{\beta R\mathbb{E}_1\left[\frac{\tilde{\theta}_2}{w\bar{H}-Rb_2}\right]}=\frac{\tilde{\theta}_1(w\bar{H}-Rb_2)}{\beta R\mathbb{E}_1\left[\tilde{\theta}_2\right]} \tag{A.15}$$

Combining (A.15) and (A.13) gives:

$$\tau_1=1-\frac{1}{\beta\mathbb{E}_1\left[\tilde{\theta}_2\right]}\left(\frac{1}{R}-\frac{b_2}{w\bar{H}}\right) \tag{A.16}$$

Replace (A.15) and (A.16) into the budget constraint of the government to obtain the law of motion of debt:

$$b_2 = \frac{w\tilde{H}\left(\frac{1+\tilde{\theta}_1}{\beta R\mathbb{E}_1[\tilde{\theta}_2]} - 1\right) + Rb_1}{1 + \frac{1+\tilde{\theta}_1}{\beta\mathbb{E}_1[\tilde{\theta}_2]}} \tag{A.17}$$

Expressing equations (A.15) to (A.17) in terms of $\overline{b}_2$ and rearranging lead to the equations in the lemma.

□

A.3 Proof of Lemma 3

Lemma 3. *Given a popularity level $\tilde{\delta}_2$, the swing voter, $\hat{H}_2(\tilde{\delta})$, is given by:*

1. *If $\omega(R) - \frac{1}{2\tilde{\Psi}} < \omega(D)$:*

$$\hat{H}_2(\tilde{\delta}) = \left(\alpha \cdot [\overline{b}_2 - b_2]^{\tilde{\theta}_D - \tilde{\theta}_R} \cdot e^{-\tilde{\delta}}\right)^{\frac{1}{\rho_D\tilde{\theta}_D - \rho_R\tilde{\theta}_R}}, \tag{A.18}$$

   where $\alpha \equiv \frac{(1+\tilde{\theta}_R)^{1+\tilde{\theta}_R}}{(1+\tilde{\theta}_D)^{1+\tilde{\theta}_D}} \frac{\tilde{\theta}_D^{\tilde{\theta}_D}}{\tilde{\theta}_R^{\tilde{\theta}_R}} (R\bar{\phi})^{\tilde{\theta}_D - \tilde{\theta}_R}$.

2. *If $\omega(R) - \frac{1}{2\tilde{\Psi}} \geq \omega(D)$: $\hat{H}_2(\tilde{\delta}) = 1$ (Republican dominance).*

Proof. Depending on $\rho_D\tilde{\theta}_D$ and $\rho_R\tilde{\theta}_R$, there are three possible cases, each one with two sub-cases:

1. **Case 1, $\rho_R\tilde{\theta}_R < 1$ and $\rho_D\tilde{\theta}_D > 1$:** $\tilde{U}_2(H|p_R)$ is increasing and $\tilde{U}_2(H|p_D)$ is decreasing
   (a) $\omega(R) - \frac{1}{2\tilde{\Psi}} < \omega(D)$: $\tilde{U}_2(H|R)$ intersects $\tilde{U}_2(H|D)$ from below at some $\hat{H}(\tilde{\delta})$ for any $\tilde{\delta} \in \left[-\frac{1}{2\tilde{\Psi}}, \frac{1}{2\tilde{\Psi}}\right]$.
   (b) $\omega(R) - \frac{1}{2\tilde{\Psi}} \geq \omega(D)$: $\tilde{U}_2(H|R)$ does not intersect $\tilde{U}_2(H|D)$. Thus, the Republican dominates for any $\tilde{\delta} \in \left[-\frac{1}{2\tilde{\Psi}}, \frac{1}{2\tilde{\Psi}}\right]$.
2. **Case 2, $\rho_R\tilde{\theta}_R < 1$ and $\rho_D\tilde{\theta}_D < 1$:** both $\tilde{U}_2(H|R)$ and $\tilde{U}_2(H|D)$ are increasing.
   (a) $\omega(R) - \frac{1}{2\tilde{\Psi}} < \omega(D)$: $\tilde{U}_2(H|R)$ intersects $\tilde{U}_2(H|D)$ from below at some $\hat{H}(\tilde{\delta})$ for any $\tilde{\delta} \in \left[-\frac{1}{2\tilde{\Psi}}, \frac{1}{2\tilde{\Psi}}\right]$.
   (b) $\omega(R) - \frac{1}{2\tilde{\Psi}} \geq \omega(D)$: $\tilde{U}_2(H|R)$ does not intersect $\tilde{U}_2(H|D)$ (since $\tilde{U}_2(H|R)$ is steeper). Thus, the Republican dominates for any $\tilde{\delta} \in \left[-\frac{1}{2\tilde{\Psi}}, \frac{1}{2\tilde{\Psi}}\right]$.
3. **Case 3, $\rho_R\tilde{\theta}_R > 1$ and $\rho_D\tilde{\theta}_D > 1$:** both $\tilde{U}_2(H|R)$ and $\tilde{U}_2(H|D)$ are decreasing.
   (a) $\omega(R) - \frac{1}{2\tilde{\Psi}} < \omega(D)$: $\tilde{U}_2(H|R)$ intersects $\tilde{U}_2(H|D)$ from below at some $\hat{H}(\tilde{\delta})$ for any $\tilde{\delta} \in \left[-\frac{1}{2\tilde{\Psi}}, \frac{1}{2\tilde{\Psi}}\right]$.
   (b) $\omega(R) - \frac{1}{2\tilde{\Psi}} \geq \omega(D)$: $\tilde{U}_2(H|R)$ does not intersect $\tilde{U}_2(H|D)$ (since $\tilde{U}_2(H|D)$ is steeper). Thus, the Republican dominates for any $\tilde{\delta} \in \left[-\frac{1}{2\tilde{\Psi}}, \frac{1}{2\tilde{\Psi}}\right]$.

Suppose that $\omega(p_R) - \frac{1}{2\tilde{\Psi}} < \omega(p_D)$ and define $\Delta\tilde{U}_2 \equiv \tilde{U}_2(H|\tilde{\theta}_D) - \tilde{U}_2(H|\tilde{\theta}_R)$. Use equation (4.11) to obtain:

$$\Delta\tilde{U}_2 = \log\left(\frac{(1+\tilde{\theta}_R)^{1+\tilde{\theta}_R}}{(1+\tilde{\theta}_D)^{1+\tilde{\theta}_D}}\frac{\tilde{\theta}_D^{\tilde{\theta}_D}}{\tilde{\theta}_R^{\tilde{\theta}_R}}(R\tilde{\phi})^{\tilde{\theta}_D-\tilde{\theta}_R}\cdot[\overline{b}_2-b_2]^{\tilde{\theta}_D-\tilde{\theta}_R}\cdot H^{p_R\tilde{\theta}_R-p_D\tilde{\theta}_D}\cdot e^{-\tilde{\delta}}\right)$$

Define $\alpha \equiv \frac{(1+\tilde{\theta}_R)^{1+\tilde{\theta}_R}}{(1+\tilde{\theta}_D)^{1+\tilde{\theta}_D}}\frac{\tilde{\theta}_D^{\tilde{\theta}_D}}{\tilde{\theta}_R^{\tilde{\theta}_R}}(R\tilde{\phi})^{\tilde{\theta}_D-\tilde{\theta}_R}$, then we obtain:

$$\Delta\tilde{U}_2 = \log\left(\alpha\cdot[\overline{b}_2-b_2]^{\tilde{\theta}_D-\tilde{\theta}_R}\cdot H^{p_R\tilde{\theta}_R-p_D\tilde{\theta}_D}\cdot e^{-\tilde{\delta}}\right) \tag{A.19}$$

The swing voter is given by imposing $\Delta\tilde{U}_2 = 0$ in equation (A.19), leading to expression (A.18).

□

A.4 Proof of Corollary 1

Corollary 1. *The minimum debt level for a young agent with productivity H to support the Republican is given by:*

$$b_2(H,\delta) \equiv \overline{b}_2 - \left(\frac{1}{\alpha}H^{\rho_D\tilde{\theta}_D-\rho_R\tilde{\theta}_R}e^{\delta}\right)^{\frac{1}{\tilde{\theta}_D-\tilde{\theta}_R}} \tag{A.20}$$

Proof. Consider an agent with labor productivity H. Then, the agent supports the Republican if:

$$\Delta\tilde{U}_2 \leq 0 \Leftrightarrow \overline{b}_2 - \left(\frac{1}{\alpha}H^{\rho_D\tilde{\theta}_D-\rho_R\tilde{\theta}_R}e^{\delta}\right)^{\frac{1}{\tilde{\theta}_D-\tilde{\theta}_R}} \leq b_2$$

Thus, the minimum inherited debt to support the Republican is given by equation (A.20).

□

A.5 Proof of Lemma 4

Lemma 4. *Suppose that the swing voter is interior, $\omega(R) - \frac{1}{2\tilde{\Psi}} < \omega(D)$. The probability that the Democrat wins the election is strictly decreasing in inherited debt b_2:*

$$\frac{\partial\Pi_{D,2}}{\partial b_2} = -\frac{(\tilde{\theta}_D-\tilde{\theta}_R)}{\overline{b}_2-b_2} < 0 \tag{A.21}$$

Proof. Using equation (3.8), the equilibrium probability that Democrat wins the election at $t=2$ is:

$$\begin{aligned}\Pi_{D,2} &= Pr\left[S_{D,2} \geq \frac{1}{2}\right] = Pr\left[\hat{H}_2(\tilde{\delta})^{-\eta} \geq \frac{1}{2}\right] \\ &= \tilde{\Psi}\left[\frac{\rho_D\tilde{\theta}_D-\rho_R\tilde{\theta}_R}{\eta}\log\left(\frac{1}{2}\right)+\log(\alpha)+(\tilde{\theta}_D-\tilde{\theta}_R)\log(\overline{b}_2-b_2)\right]\end{aligned} \tag{A.22}$$

Differentiation in terms of b_2 leads to equation (A.21). □

A.6 Proof of Lemma 5

Lemma 5. *The effect of current debt (b_1) on future debt (b_2) is given by:*

$$\frac{\partial b_2}{\partial b_1} = \frac{\beta R \mathbb{E}_1\left[\tilde{\theta}_2\right]\left(1+\tilde{\theta}_1+\beta\mathbb{E}_1\left[\tilde{\theta}_2\right]\right)}{\left(1+\tilde{\theta}_1+\beta\mathbb{E}_1\left[\tilde{\theta}_2\right]\right)^2 - \frac{\beta(\tilde{\theta}_D-\tilde{\theta}_R)^2(1+\tilde{\theta}_1)(2\overline{b}_2-b_1)}{\overline{b}_2-b_2}} > 0 \tag{A.23}$$

Proof. Using equation (A.26), implicit differentiation of (A.7) gives:

$$\frac{\partial b_2}{\partial b_1} = \frac{\beta R \mathbb{E}_1\left[\tilde{\theta}_2\right]\left(1+\tilde{\theta}_1+\beta\mathbb{E}_1\left[\tilde{\theta}_2\right]\right)}{\left(1+\tilde{\theta}_1+\beta\mathbb{E}_1\left[\tilde{\theta}_2\right]\right)^2 + \beta\frac{\partial\mathbb{E}_1[\tilde{\theta}_2]}{\partial b_2}(1+\tilde{\theta}_1)(2\overline{b}_2-b_1)} \tag{A.24}$$

The expected value of $\tilde{\theta}_2$ is given by:

$$\mathbb{E}_1\left[\tilde{\theta}_2\right] = \tilde{\theta}_R + (\tilde{\theta}_D-\tilde{\theta}_R)\Pi_{D,2}, \tag{A.25}$$

which is strictly decreasing in inherited debt:

$$\frac{\partial\mathbb{E}_1\left[\tilde{\theta}_2\right]}{\partial b_2} = (\tilde{\theta}_D-\tilde{\theta}_R)\frac{\partial\Pi_{D,2}}{\partial b_2} = -\frac{(\tilde{\theta}_D-\tilde{\theta}_R)^2}{\overline{b}_2-b_2} < 0 \tag{A.26}$$

Replacing (A.26) in (A.24) leads to expression (A.23).

□

A.7 Proof of Lemma 6

Lemma 6. *In a stationary Markov-perfect equilibrium, the policy functions of the type-p party are*

$$b_{t+1}(b_t \mid p) = \bar{b} - \frac{R\beta\mathbb{E}_t[\tilde{\theta}_{t+1}]\left(\bar{b}-b_t\right)}{1+\tilde{\theta}_p+\beta\mathbb{E}_t[\tilde{\theta}_{t+1}]}, \tag{A.27}$$

$$\tau(b_t \mid p) = 1 - \frac{1}{(R-1)\beta\mathbb{E}_t[\tilde{\theta}_{t+1}]}\left(1-\frac{b_{t+1}(b_t \mid p)}{\bar{b}}\right), \tag{A.28}$$

$$g(b_t \mid p) = \frac{R\tilde{\theta}_p\left(\bar{b}-b_t\right)}{1+\tilde{\theta}_p+\beta\mathbb{E}_t[\tilde{\theta}_{t+1}]}. \tag{A.29}$$

Proof. We guess that the equilibrium public good policy function is linear in the fiscal space ($\bar{b}-b$),

$$g(b \mid p) = \gamma_p\left(\bar{b}-b\right), \tag{A.30}$$

with $\gamma_p > 0$ a constant to be determined. Under (A.30),

$$\log g_{t+1} \;=\; \log\gamma_{p_{t+1}} + \log\big(\bar{b} - b_{t+1}\big), \quad \text{so} \quad \frac{\partial \log g_{t+1}}{\partial b_{t+1}} \;=\; -\,\frac{1}{\bar{b} - b_{t+1}}, \tag{A.31}$$

which is independent of $\gamma_{p_{t+1}}$ and of the type p_{t+1}. Hence, when we take the expectation over p_{t+1} in the FOC for b_{t+1}, the only surviving forward-looking object is the scalar $\mathbb{E}_t[\tilde{\theta}_{t+1}]$.

Step 1: First-order conditions. Substituting the budget constraint for g_t in the maximization problem (5.6), the FOC with respect to τ_t uses $\partial v_t/\partial\tau_t = -w$ and $\partial g_t/\partial\tau_t = w\bar{H}$:

$$\frac{1}{v_t}\frac{\partial v_t}{\partial \tau_t} + \frac{\tilde{\theta}_p}{g_t}\frac{\partial g_t}{\partial \tau_t} = 0 \iff g_t = \tilde{\theta}_p\,(1-\tau_t)\,w\bar{H}. \tag{A.32}$$

The FOC with respect to b_{t+1}, using $\partial g_t/\partial b_{t+1} = 1$ and (A.31), is

$$\frac{\tilde{\theta}_p}{g_t} \;+\; \beta\,\mathbb{E}_t\left[\tilde{\theta}_{t+1}\cdot\left(-\frac{1}{\bar{b} - b_{t+1}}\right)\right] \;=\; 0. \tag{A.33}$$

Because b_{t+1} is chosen at t, $(\bar{b} - b_{t+1})$ is measurable with respect to t-information and comes out of the expectation, leaving the generalized Euler equation

$$\frac{\tilde{\theta}_p}{g_t} \;=\; \frac{\beta\,\mathbb{E}_t[\tilde{\theta}_{t+1}]}{\bar{b} - b_{t+1}}. \tag{A.34}$$

Rearranging,

$$g_t \;=\; \frac{\tilde{\theta}_p\,\big(\bar{b} - b_{t+1}\big)}{\beta\,\mathbb{E}_t[\tilde{\theta}_{t+1}]}. \tag{A.35}$$

Step 2: Solving for b_{t+1}. Equate (A.35) with (A.32) and use $w\bar{H} = (R-1)\bar{b}$:

$$\tilde{\theta}_p\,(1-\tau_t)\,(R-1)\,\bar{b} \;=\; \frac{\tilde{\theta}_p\,(\bar{b} - b_{t+1})}{\beta\,\mathbb{E}_t[\tilde{\theta}_{t+1}]}. \tag{A.36}$$

Solving (A.36) for $1-\tau_t$ and substituting into the budget constraint of the government written as $\tau_t(R-1)\bar{b} = g_t + Rb_t - b_{t+1}$, then eliminating g_t via (A.35), gives

$$(R-1)\bar{b} \;-\; \frac{\bar{b} - b_{t+1}}{\beta\,\mathbb{E}_t[\tilde{\theta}_{t+1}]} \;=\; \frac{\tilde{\theta}_p(\bar{b} - b_{t+1})}{\beta\,\mathbb{E}_t[\tilde{\theta}_{t+1}]} + Rb_t - b_{t+1}. \tag{A.37}$$

Multiplying through by $\beta\,\mathbb{E}_t[\tilde{\theta}_{t+1}]$ and using $(R-1)\bar{b} - Rb_t + b_{t+1} = R(\bar{b} - b_t) - (\bar{b} - b_{t+1})$, this rearranges to

$$\big(1 + \tilde{\theta}_p + \beta\,\mathbb{E}_t[\tilde{\theta}_{t+1}]\big)\big(\bar{b} - b_{t+1}\big) \;=\; R\,\beta\,\mathbb{E}_t[\tilde{\theta}_{t+1}]\,\big(\bar{b} - b_t\big), \tag{A.38}$$

which is (5.7).

Step 3: Closed forms for g_t and τ_t. Substitute (A.38) into (A.35): the factor $(\bar{b} - b_{t+1})$ on the right-hand

side of (A.35) is replaced by $R\beta\mathbb{E}_t[\tilde{\theta}_{t+1}]\,(\bar{b}-b_t)/\big(1+\tilde{\theta}_p+\beta\mathbb{E}_t[\tilde{\theta}_{t+1}]\big)$, giving

$$g(b_t \mid p) \;=\; \frac{R\tilde{\theta}_p\,(\bar{b}-b_t)}{1+\tilde{\theta}_p+\beta\mathbb{E}_t[\tilde{\theta}_{t+1}]}, \tag{A.39}$$

which is (5.9). Solving (A.36) for τ_t,

$$\tau_t \;=\; 1 \;-\; \frac{\bar{b}-b_{t+1}}{(R-1)\,\bar{b}\,\beta\mathbb{E}_t[\tilde{\theta}_{t+1}]} \;=\; 1 \;-\; \frac{1}{(R-1)\,\beta\mathbb{E}_t[\tilde{\theta}_{t+1}]}\left(1-\frac{b_{t+1}}{\bar{b}}\right), \tag{A.40}$$

which is (5.8).

Step 4: Verification of the guess. From (A.39), the equilibrium public good policy function is exactly of the form (A.30) with coefficient

$$\gamma_p \;=\; \frac{R\tilde{\theta}_p}{1+\tilde{\theta}_p+\beta\mathbb{E}_t[\tilde{\theta}_{t+1}]} \;>\; 0, \tag{A.41}$$

confirming the guess. Hence (5.7)–(5.9) are the equilibrium policy functions. □

A.8 Proof of Proposition 1

Proposition 1.

1. *There exists a unique party-specific stationary public debt level, $b_{t+1}(b_p^*|p)=b_p^*$, with $b_D^*>b_R^*$.*
2. *The transition dynamics of public debt and electoral uncertainty are given by:*
   - *(a) If $b_t<b_R^*$, then b_{t+1} follows a stochastic increasing path, where $b_{t+1}=b_{t+1}(b_t|D)$ with probability $\Pi_{D,t}$ and $b_{t+1}=b_{t+1}(b_t|R)$ with probability $\Pi_{R,t}$. Moreover, $\Pi_{R,t}$ increases over time as debt approaches to b_R^*.*
   - *(b) If $b_t\in[b_R^*,b_D^*]$, public debt attains a stochastic-cycling equilibrium, where b_t stays in $[b_R^*,b_D^*]$ forever. Debt increases to $b_{t+1}=b_{t+1}(b_t|D)$ with probability $\Pi_{D,t}$, while it decreases to $b_{t+1}=b_{t+1}(b_t|R)$ with probability $\Pi_{R,t}$. Moreover, $\Pi_{R,t}$ increases when debt approaches to b_D^*, while $\Pi_{R,t}$ decreases when debt approaches to b_R^*.*
   - *(c) If $b_t>b_D^*$, then b_{t+1} follows a stochastic decreasing path, where $b_{t+1}=b_{t+1}(b_t|D)$ with probability $\Pi_{D,t}$ and $b_{t+1}=b_{t+1}(b_t|R)$ with probability $\Pi_{R,t}$. Moreover, $\Pi_{R,t}$ decreases over time as debt approaches to b_R^*.*

Proof. We proof the result for the two-period model, which can be extended to the infinite period model. Fix $p\in\{D,R\}$. The debt policy function takes the form

$$b_{t+1}(b\mid p)=\frac{\bar{b}(1+\widetilde{\theta}_p)-\beta E_t[\widetilde{\theta}_{t+1}](\bar{b}-b)}{1+\widetilde{\theta}_p+\beta E_t[\widetilde{\theta}_{t+1}]}. \tag{A.42}$$

Part 1: Existence and uniqueness of b_p^*. Imposing $b_{t+1}(b \mid p) = b$ in (A.42) and simplifying, a stationary debt level is a zero of

$$\Lambda_p(b) \equiv \bar{b}\left[1 - \frac{\beta E_t[\widetilde{\theta}_{t+1}]}{1+\widetilde{\theta}_p}\right] - b. \tag{A.43}$$

Differentiating (A.43) twice and applying Lemma 4,

$$\Lambda_p''(b) = \frac{\beta\bar{b}(\widetilde{\theta}_D - \widetilde{\theta}_R)^2}{(1+\widetilde{\theta}_p)(\bar{b}-b)^2} > 0,$$

so Λ_p is strictly convex on $(0, \bar{b})$. At the boundaries, $\Lambda_p(0) > 0$ under the maintained restriction $1 + \widetilde{\theta}_p > \beta E_t[\widetilde{\theta}_{t+1} \mid b = 0]$, while

$$\Lambda_p(\bar{b}) = -\frac{\beta\bar{b}\, E_t[\widetilde{\theta}_{t+1} \mid b = \bar{b}]}{1+\widetilde{\theta}_p} < 0.$$

By the intermediate value theorem Λ_p has at least one zero in $(0, \bar{b})$, and strict convexity implies at most one: since $\Lambda_p(\bar{b}) < 0$, the function cannot return positive after any crossing. Hence, b_p^* exists and is unique for each party p.

Part 2: $b_D^* > b_R^*$. Since $\widetilde{\theta}_D > \widetilde{\theta}_R$, we have $\Lambda_D(b) > \Lambda_R(b)$ for all $b \in (0, \bar{b})$. Evaluating at b_R^* gives $\Lambda_D(b_R^*) > \Lambda_R(b_R^*) = 0$. Since $\Lambda_D(\bar{b}) < 0$ and Λ_D has a unique zero, $b_D^* > b_R^*$. This proves Part 1 of the proposition.

Part 3: Transition dynamics. Subtracting b from (A.42) and simplifying,

$$b_{t+1}(b \mid p) - b = \frac{(\bar{b}-b)(1+\widetilde{\theta}_p) - \beta\bar{b}\, E_t[\widetilde{\theta}_{t+1}]}{1+\widetilde{\theta}_p + \beta E_t[\widetilde{\theta}_{t+1}]}. \tag{A.44}$$

Since the denominator is positive, $\operatorname{sgn}(b_{t+1} - b) = \operatorname{sgn}(\Lambda_p(b))$. Combined with the sign pattern of Λ_p,

$$b_{t+1}(b \mid p) > b \iff \Lambda_p(b) > 0 \iff b < b_p^*, \tag{A.45}$$

$$b_{t+1}(b \mid p) < b \iff \Lambda_p(b) < 0 \iff b > b_p^*. \tag{A.46}$$

Case 2(a): $b_t < b_R^* < b_D^*$. Both $\Lambda_R(b_t) > 0$ and $\Lambda_D(b_t) > 0$, so debt rises regardless of the winner. By Lemma 4, $\partial\Pi_{R,t}/\partial b_t > 0$, so rising debt raises the probability of a Republican win. This proves Part 2(a).

Case 2(b): $b_t \in [b_R^*, b_D^*]$. From (A.45)–(A.46), $b_{t+1}(b_t \mid R) \le b_t \le b_{t+1}(b_t \mid D)$: the Republican decreases debt and the Democrat increases it. Since the policy function is increasing in b_t (Lemma 5) and each stationary level is a fixed point, $b_{t+1}(b_t \mid R) \ge b_{t+1}(b_R^* \mid R) = b_R^*$ and $b_{t+1}(b_t \mid D) \le b_{t+1}(b_D^* \mid D) = b_D^*$, so debt stays in $[b_R^*, b_D^*]$ forever. Since $\partial\Pi_{R,t}/\partial b_t > 0$, the probability of a Republican win is increasing in debt. This proves Part 2(b).

Case 2(c): $b_t > b_D^* > b_R^*$. Both $\Lambda_D(b_t) < 0$ and $\Lambda_R(b_t) < 0$, so debt falls regardless of the winner. Since $\partial\Pi_{R,t}/\partial b_t > 0$, falling debt reduces the probability of a Republican win. This proves Part 2(c) and completes the proof. □

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B Data Construction

B.1 Data collection

This section explains how the data presented in the main text was constructed. We list below the sources used to build the election-party panel, describe the criteria for classifying parties as extreme, and document every adjustment made to the raw data on a country-by-country basis.

The primary source for parliamentary elections is the ParlGov database, which provides party-level vote shares, seat counts, and cabinet composition for all democratic elections in our sample countries since 1900. ParlGov assigns each party a family classification code (e.g., `com` for Communist/Socialist, `right` for Right-wing, `soc` for Social Democracy, `lib` for Liberal, `con` for Conservative, `agr` for Agrarian, among others). Following Funke et al. (2016), we classify parties belonging to the Communist/Socialist family (`com`) as extreme left and those belonging to the Right-wing family (`right`) as extreme right. Where ParlGov's classification diverges from that of Funke et al. (2016) or the academic literature, we recode accordingly; each such change is documented below.

Cabinet status is determined by merging ParlGov's cabinet data with the election-party file. Multiple cabinets may form after a single election (e.g., a caretaker government followed by a permanent coalition). We merge on election and party identifiers and select the first non-caretaker cabinet formed after each election. For elections in which all cabinets are caretaker governments, we use the first cabinet. Parties are coded as government if they hold ministerial seats in the resulting cabinet, as opposition if they do not, and as missing if they provided external legislative support without joining the cabinet (e.g., Denmark 2015).

Debt-to-GDP ratios are constructed by combining the IMF Historical Public Debt Database (HPDD), which covers the period 1800–2015, with the IMF Fiscal Monitor, which extends the series through 2024. The HPDD serves as the base series; observations from the Fiscal Monitor fill the years 2016–2024. GDP data come from the Maddison Project Database (up to 2022), extrapolated to 2023 using real GDP growth rates from the IMF World Economic Outlook.

The sample covers 37 countries: Australia, Austria, Belgium, Bulgaria, Canada, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Israel, Italy, Japan, Latvia, Lithuania, Luxembourg, Malta, Netherlands, New Zealand, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkey, and the United Kingdom. Cyprus is excluded from the government–opposition margin analysis because it has a presidential regime, and Switzerland is excluded from this variable because it has maintained a balanced federal government with all major parties in coalition since 1943, making the notion of a government–opposition margin inapplicable.

The remainder of this section documents, country by country, the specific adjustments made to party ideology classifications, vote share corrections, and cabinet status recodings.

B.2 Party ideology reclassifications

The following reclassifications were applied to the ParlGov family codes to ensure consistency with the classification scheme used in Funke et al. (2016).

Australia The Nationalist Party of Australia is reclassified from `right` to `con`. Williams (1971) documents that this party should be classified as mainstream conservative rather than extreme right.

Belgium The Libertarian-Direct-Democratic Party is reclassified from `lib` to `right`, following Mudde and Rovira Kaltwasser (2012), who identify it as a right-wing populist formation.

Denmark The Progress Party is reclassified from `lib` to `right`, as documented by Hainsworth (2008) and Topaloff (2012). Additionally, the Socialist People's Party is reclassified to `com`, following Funke et al. (2016).

Finland The Finnish Party (True Finns) is reclassified from `agr` to `right`. Arter (2010) documents its breakthrough as a populist radical right party.

France Three reclassifications are applied. The Republican Union is reclassified from `right` to `con`, following Le Béguec (2007). The Movement for France is reclassified from `con` to `right`, as Libération (2005) identifies it as a shelter for the extreme right. The Citizens' Movement is reclassified from `com` to `soc`, following Gougou and Labouret (2010), who classify it as social-democratic rather than extreme left.

Italy The National Alliance is reclassified from `con` to `right`, in accordance with Catellani et al. (2005), who document its post-fascist extreme-right character. The Communist Refoundation Party is reclassified from `soc` to `com`, following Funke et al. (2016).

Switzerland The Swiss People's Party is reclassified from `agr` to `right`, given its anti-immigration platform and populist radical right orientation (The Guardian, 2023).

United Kingdom The Respect – The Unity Coalition is reclassified from `code` to `com`, following Benedek (2007) and Clark (2012), who identify it as a leftist-Islamist alliance.

B.3 Cabinet status corrections and vote share adjustments

Below we list the corrections applied to cabinet status assignments and vote shares. These address cases where ParlGov's automated cabinet merge yields incorrect assignments or where vote shares in the database diverge from official electoral records.

Austria In 1959, 1962, 1971, 1975, and 1986, the Communist Party (KPÖ) is coded as opposition. In 1962, the caretaker flag is additionally set for the KPÖ. In 1966, the Progressive Democrats (DFP) are coded as opposition. In 1970, the Freedom Party (FPÖ) is coded as government and the KPÖ as opposition. In 1983, the Alternative List (ALÖ) and the United Greens (VGÖ) are coded as opposition. In 1990, the VGÖ is coded as opposition. In 1999, the Social Democrats (SPÖ) are coded as opposition, the FPÖ as government, and the Liberal Forum (LIF) as opposition. In 2002 and 2008, the LIF is coded as opposition; in 2008, the Citizens' Forum Austria (Fritz) is also coded as opposition. In 2006, the KPÖ and the Hans-Peter Martin List are coded as opposition. In 2013, the KPÖ and the Alliance for the Future of Austria (BZÖ) are coded as opposition. In 2017, the Greens (Grüne) and My Vote Counts! (GILT) are coded as opposition. In 2019, the Grüne and the Austrian People's Party (ÖVP) are coded as government, while

Liste Pilz is coded as opposition.

Belgium In 1925, the Liberal Party (LP-PL) is coded as opposition. In 2014, the Socialist Party (PS), the Francophone Socialists (SP), and the Francophone Christian Social Party (PSC-CDH) are coded as opposition, while the New Flemish Alliance (N-VA) is coded as government (Inter-Parliamentary Union, 2014).

Bulgaria The elections of 2021 and 2022 are excluded from the analysis due to political instability that prevented the formation of stable governments.

Canada In 1953, the Liberal Party vote share is corrected to 48.43%, in accordance with official records (Government of Canada, n.d.). In 2004 and 2008, the Green Party (GPC) is coded as opposition. In 2019 and 2021, the People's Party of Canada (PPC) is coded as opposition.

Cyprus Cyprus is excluded from the government–opposition margin analysis entirely because it operates under a presidential regime.

Denmark In 1973, five parties that supported the minority government without joining the cabinet—the Progress Party (FrP), the Conservative People's Party (KF), the Social Liberal Party (RV), the Centre Democrats (CD), and the Christian People's Party (KrF)—are coded as missing (Skjæveland, 2009). In 2015, the Conservative People's Party, the Danish People's Party (DF), and the Liberal Alliance (NLA) are coded as missing, as they supported the government without holding cabinet seats (The Local, 2015). In 2019, the Social Liberal Party (RV) and the Socialist People's Party (SF) are coded as government (The Local, 2019).

Finland In 1939, the National Coalition Party (KOK) and the Swedish People's Party (RKP-SFP) are coded as opposition (Finnish Government, n.d.). In 1970, the Social Democratic Party (SSDP), the Centre Party (KESK), the Liberal Party (KE/SLK), and the Swedish People's Party (RKP-SFP) are coded as government.

France The 1945 election is excluded because France was under a provisional government. In 1946, the Rally of the Left (RGR) is coded as opposition. In 1951, the RGR and the Socialist Party (PS) are coded as government. In 1958, the Radical Socialist Party (PR) and the Popular Republican Movement (MRP) are coded as opposition. In 1978, the vote share of the Union for French Democracy (UDF) is corrected to 21.45%. In 1981, the UDF vote share is corrected to 19.13%, and the Greens (V) are coded as opposition. In 1986, the RPR and UDF coalition is coded as government and the Greens as opposition. In 1988, the Left Radical Party (PRG) and the Presidential Majority (MP) are coded as government. In 2012, the French Communist Party (PCF) is coded as missing because it provided external support to the government without holding cabinet seats. In 2017, the Republicans (LR) are coded as opposition (Ortiz de Zárate Arce, 2024; France-Politique, n.d.; Macridis and Brown, 1960).

Germany The 1932 election is adjusted so that the Nazi Party (NSDAP) is coded as missing, reflecting an implicit agreement with Hitler rather than a formal coalition. The 1933 election is excluded entirely as it was not conducted under democratic conditions (DW/Deutsche Welle, 2017). In all subsequent elections, the CDU and CSU are coded consistently: as government in 1949, 1953, 1957, 1965, 1983, 1987, 1994, 2005, 2009, 2013, and 2017; and as opposition in 1969, 1972, 1976, 1980, 1998, 2002, and 2021. The Communist Party (KPD) is coded as opposition in 1953. The SPD and FDP are coded as opposition in

1957. The All-German Party (GDP) and the German Peace Union (DFU) are coded as opposition in 1961; the DFU is also coded as opposition in 1965. The National Democratic Party (NPD) is coded as opposition in every election from 1965 through 2013 in which it competed. The Alliance 90/Greens (B90/Grü) is coded as opposition in 1980. The Republicans (Rep) are coded as opposition in 1994, 1998, and 2002. The German People's Union (DVU) is coded as opposition in 1998. The Alternative for Germany (AfD) is coded as opposition in 2013.

Hungary In 2006, the Fidesz–KDNP alliance is coded as opposition. In 2010, 2014, and 2018, the same alliance is coded as government.

Iceland In 1942, the Independence Party (Sj) is coded as government.

Ireland In 2011, the Green Party is coded as opposition, and the observation for candidates with no party affiliation is coded as missing. In 2020, Fianna Fáil (FF) is coded as government (Irish Times, 2016; RTÉ, 2022).

Israel In 1949, Mapai (Ma/Av), the United Religious Front (HDM), the Progressive Party (MP), and the Sephardi and Oriental Communities list (SVM) are coded as government.

Italy In 1994, vote shares are corrected to match official records from Ministero dell'Interno (2008): the Federation of the Greens (FdV) to 2.7%, Forza Italia (FI-PdL) to 21.01%, the National Alliance (AN) to 13.47%, the Northern League (LN) to 8.36%, and the Democrats of the Left (DS) to 20.36%. In 1996, vote shares are similarly corrected: FdV to 2.5%, DS to 21.06%, Forza Italia to 20.57%, National Alliance to 15.66%, Popolari per Prodi (PpP) to 6.81%, the Christian Democratic Centre (CCD+CDU) to 5.84%, and Italian Renewal (DINI-RI) to 4.34%. In 2013, the Communist Refoundation Party (PRC) and Fare per Fermare il Declino (FiD) are coded as opposition.

Japan In 1952, the Japanese Communist Party (JCP) is coded as opposition. For Japan, we use representative vote shares throughout, as the original series reports constituency-level vote totals.

Luxembourg In 1922, the Party of the Right (PD) is coded as government and the National Independent Party (ONP) as opposition. In 1925, the Radical Party (RP) is coded as government (Gouvernement du Grand-Duché de Luxembourg, 2011).

Netherlands In 1977, the Labour Party (PvdA) is coded as opposition. In 2017, the Christian Democratic Appeal (CDA), Democrats 66 (D66), and the Christian Union (CU) are coded as government, while the PvdA is coded as opposition.

New Zealand In 1925, the Reform Party (RP) is coded as government and the Liberal Party as opposition. In 1931, the Reform Party is coded as government (Bassett, 1982; Sinclair and Dalziel, 2000).

Norway In 1903, the Conservative Party (Høyre) is coded as opposition (The Government of Norway, n.d.).

Portugal The 1975 election is excluded because no government was elected. In 1979 and 1980, the Democratic Alliance (AD) is coded as government and the United People's Alliance (APU) as opposition (Governo da República Portuguesa, n.d.).

Sweden The 1914 election is excluded because Sweden was under a non-parliamentary government at the time (Hadenius, 2003). In 1920, the Moderate Party (M) is coded as government.

Switzerland Switzerland has maintained a balanced federal government since 1943 in which the

four largest parties share executive power. This makes the concept of a government–opposition victory margin inapplicable. Accordingly, cabinet status variables are set to missing for all Swiss elections, and Switzerland is excluded from the margin of victory analysis.

United Kingdom Vote shares for the Conservative Party are corrected to match official records in the following elections: 1950 (43.4%), 1951 (48.0%), 1955 (49.7%), 1959 (49.4%), 1964 (43.4%), and 1966 (41.9%). In 1935, the Labour Party is coded as opposition. In 1983, the Liberal vote is aggregated correctly rather than being disaggregated as in the original series (Cracknell et al., 2023).

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